Aspirations and Hope of Smallholders Regarding Financial Well-Being in Old Age: Evidence from an RCT in Rural Ghana *

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Abstract
This study evaluates two randomized interventions designed to encourage smallholder cocoa farmers in Ghana to save for old age: a video featuring peer testimonies, intended to raise aspirations for financial well-being in old age, and a "savings calendar," intended to increase the salience of savings needs and specific savings goals. We find that the video significantly enhances aspirations in the short run. In the long run, the video combined with the savings calendar significantly increases the fraction of farmers saving for old age, with effects on pension savings amounts concentrated among farmers with higher initial total savings. Long-run effects on aspirations are unevenly distributed: the marginal effect of the video combined with the savings calendar is smaller for women, while the marginal effects of the video, with or without the calendar, are larger among farmers with higher initial savings. Our results inform the design of pro-poor old-age welfare policies in sub-Saharan Africa against the backdrop of rising life expectancies and weak formal safety nets.

Keywords: Aspirations; Hope; Pension Savings; Randomized Control Trial; Video

JEL Classification: D91; O12; O16; G51

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1 Introduction

Poverty among older adults is an increasingly serious challenge in many developing countries (e.g., Banerjee et al., 2024), especially those in sub-Saharan Africa (SSA). SSA has recorded the world's largest gains in life expectancy over the past decade and is projected to continue doing so until the 2040s; consequently, the region's older (65+) population is expected to rise from around 41 million in 2025 to roughly 102 million by 2050 (United Nations, 2024). At the same time, traditional social security systems, such as family- and community-based safety nets, are eroding (Duhon et al., 2025; Kos and Lensink, 2023).

In light of these developments, pension schemes and other long-term savings mechanisms aimed at enhancing the well-being of older adults are crucial (Aboderin, 2017; Huang and Zhang, 2021). Yet, although several countries in the region have introduced public social protection programs, including social pensions, and the private sector started offering a variety of pension products as well, pension coverage remains alarmingly low — at only 19.8% in sharp contrast to the global average of 77.5% (International Labour Organization, 2021). Among the poor, uptake and use of long-term savings accounts also remain very limited. For smallholders, who face large income fluctuations and are especially vulnerable to shocks, accumulating any type of long-term savings is particularly challenging.

Research on micro-savings products — financial tools explicitly tailored to help low-income individuals, including smallholders — shows that removing external barriers to accessing savings accounts, such as high account-opening fees or complicated procedures, significantly increases uptake. This indicates a high latent demand for formal savings products among the poor (e.g., Dupas and Robinson, 2013; Karlan et al., 2014; Pomeranz and Kast, 2024). However, uptake does not automatically translate into usage: Numerous empirical studies report mixed findings even for short-term usage rates (Dupas and Robinson, 2013; Prina, 2015; Schaner, 2017; Dupas et al., 2018), and institutions such as the Center for Financial Inclusion have raised concerns about high dormancy rates, particularly in South Asia and SSA (Rhyne and Kelly, 2018). This suggests that *external* barriers are only part of the problem: Interventions that tackle them may have limited effects when *internal*, psychological constraints are present. In particular, theory highlights how low aspirations and hope can generate behavioral poverty traps (Dalton et al., 2016; Genicot and Ray, 2017). Building on these insights, a growing experimental literature examines lifting aspirations and strengthening hope as pathways out of poverty (Beaman et al., 2012; Bernard et al., 2019, 2023; McKenzie et al., 2022).

Against this backdrop, we present results from a randomized controlled trial (RCT) in rural Ghana that tests two behavioral interventions — a video and a "savings calendar" — designed to encourage smallholder cocoa farmers to save more for old age. Our video focuses on positive testimonies from cocoa farmers who saved toward old age and realized their aspirations. Our savings calendar is an annual wall calendar that displays prominent motifs suggestive of cocoa farming and savings accumulation, in which those who received it were asked to write in the blank spaces of the calendar: (i) one long-term non-monetary savings goal for old age (e.g., owning a house) and (ii) a monthly savings amount toward achieving the stated goal.

The video seeks to raise aspirations and hope regarding financial well-being later in life, while the savings calendar is intended to make the need to save for old age and farmers' specific savings goals for old age more salient, aiming to sustain any effects of the video over time. We examine the impact of these treatments on smallholders' aspirations and hope, as well as on their (pension) savings.

Our first intervention — the video — is grounded in the literature on aspirations and hope. The literature on aspirations originates in the field of positive psychology, which focuses on factors that enhance human well-being (Seligman and Csikszentmihalyi, 2000). In particular, our video intervention draws on Ray's (2006) model, which introduces the *aspirations window*, i.e., the set of people one perceives as similar to oneself who are used as reference points when forming aspirations.[1] Ray argues that future-oriented behavior is shaped by the *aspirations gap*: the difference between the living standard a person aspires to (defined by their aspirations window) and their current standard. According to him, people minimize a weighted average of this gap and the cost of investment needed to close it. A key result from Ray's analysis is that very small or very large aspiration gaps can lead to minimal investment, a phenomenon Ray calls *aspirations failure*. In short, aspirations lead to tangible changes only when the window is open but not too wide, i.e., when relevant role models exist but do not seem too far removed, as with the cocoa farmers in our sample who watched video testimonies from peers who had saved successfully. Building on Ray (2006), Dalton et al. (2016) argue that persistent aspirations failure can result in a self-perpetuating cycle of low aspirations and low living standards; interventions that raise aspirations may therefore be required to break the cycle.

Indeed, in agriculture, higher aspirations — often fostered by targeted interventions — have

[1] Bandura (1977) describes this as the power of vicarious experience — seeing similar others succeed increases aspirations.

been found to be linked to greater technology adoption and investment and, in some studies, higher short-term savings (Nandi and Nedumaran, 2021; Bernard et al., 2023). Whether such aspiration gains translate into increased saving for old age, however, has yet to be tested — a gap this study aims to fill.

The concept of aspirations is closely related to the concept of hope. In his seminal work, Snyder (2000) defines hope as *"a positive motivational state that is based on an interactively derived sense of successful (a) agency (goal-directed energy) and (b) pathways (planning to meet goals)"*. In other words, hope consists of two elements: (1) *agency*, i.e., the motivation or determination to pursue one's aspirations, and (2) *pathways*, which refer to one's perceived ability to plan different routes that can be taken to meet those aspirations. Building on Snyder (2000), Lybbert and Wydick (2018) coined the term *aspirational hope*, combining aspirations with Snyder's hope framework of agency and pathways. Accordingly, this study examines effects on agency and pathways in addition to aspirations, as well as on (pension) savings. In line with Snyder's (2000) conceptualization, and consistent with much of the theoretical literature (see also Martin, 2014), which views hope as directed toward a desired outcome, we expect the sub-indices of hope to be affected by the intervention only after individuals revise their aspirations.

A comprehensive review of hope and economic outcomes in low- and middle-income settings (Lybbert and Wydick, 2022) finds links to asset accumulation, but identifies no evidence on pension saving. Only one study — an RCT with low-income mothers in the United States — links higher hope to larger voluntary retirement contributions (White et al., 2020). Thus, whether fostering hope can boost old-age saving among the poor — including the rural poor, most importantly smallholders — in developing countries remains untested, a gap this paper addresses.

Our calendar intervention is informed by two strands of evidence. First, the literature on aspirations interventions suggests that while short-term aspiration boosts can persist (e.g., Orkin et al., 2023), their magnitude may taper (Bernard et al., 2023). Second, the savings literature underscores the importance of making savings goals salient. For example, Karlan et al. (2016) show that reminders can increase savings by making savings goals more salient, serving as a nudge, and helping individuals stay committed to their plans and take action toward their goals. Similarly, Soman and Zhao (2011) find that visual reminders that make savings goals more salient improve goal attainment.

Taken together, these insights led us to adopt a calendar that, by making the savings goals visible every day, keeps the video-induced aspirations salient, aiming to sustain any potential effects of the video intervention over the long term and ultimately improve (pension) savings.

We estimate the effect of our interventions using a sample of smallholder cocoa farmers in the Fanteakwa and Suhum cocoa districts in Ghana's Eastern Region. This focus on Ghanaian smallholder cocoa farmers is motivated by the following considerations. First, their savings constraints are well documented: like other smallholders, smallholder cocoa farmers face seasonal income spikes, long lean periods, and high vulnerability to shocks (Kos and Lensink, 2023), leaving few able to build reserves for old-age security. Second, they embody a region-wide demographic pattern. Across SSA, farmers are significantly older than the average person, with roughly 60 percent of Africans being under 24 years of age.

We collaborated with People's Pension Trust (PPT), a licensed pension administrator for both low-income formal workers and informal workers in Ghana. We randomly assigned 956 cocoa farmers, all of whom had existing pension savings accounts with PPT (i.e., farmers with less of a risk of binding external constraints to saving for old age), to two treatment groups (one combining the video with a regular "placebo" calendar, the other combining the video with the savings calendar) and a control group (that only received the placebo calendar and did not see the video), stratified at the community level. The calendar intervention was implemented after the baseline, which included the video intervention at its end for those assigned to either of the two treatment groups, and after a subsequent remeasurement of aspirations (midline).

Our main outcome variables are survey-based measures of aspirations and hope, as well as (pension) savings. We measured aspirations at three points: at baseline (i.e., before the video intervention), at midline (i.e., immediately after the video intervention), and at endline (i.e., 12 months after the baseline). Hope, i.e., agency and pathways, was measured at baseline and again at endline.[2]

Besides estimating the average impact of our interventions on aspirations, agency and pathways, we also conduct heterogeneity analysis to examine how these effects vary for gender and initial savings level. First, regarding gender, women provide a large share of farm labor, yet lack secure land rights, access to credit, and decision-making power (Nandi and Nedumaran, 2021). More broadly, women in agrarian economies frequently lack social and economic power (Ashraf, 2009; Beaman et al., 2012; Duflo, 2012; Sharaunga et al., 2016). As a result, the ef-

[2]Note that when discussing our results below, we refer to midline and endline outcomes as short-term and long-term, respectively.

fectiveness of interventions aimed at enhancing aspirations may be more limited for women. Second, we examine heterogeneity by initial savings levels. This investigation is informed by related literature (Gehrke et al., 2023), which finds that treatment effects on educational aspirations are higher for those who already received better grades at baseline.

We find a significant positive effect of our video on aspirations in the short-term. While there are no significant long-term average treatment effects of the two treatments on aspirations, we find heterogeneity along two dimensions. First, the marginal effect of the video in combination with the savings calendar on aspirations is lower for females; though, overall, females are not worse off as a result of the treatment compared to females who did not receive the treatment. Second, higher initial levels of total savings predict higher marginal effects of the video in combination with either the placebo or the savings calendar on aspirations, a finding reminiscent of the findings of Gehrke et al. (2023) regarding educational outcomes.

We also find a positive long-run average treatment effect of the video in combination with the placebo calendar on farmers' hope, driven by an increase in agency. Even though the corresponding average treatment effect for the video in combination with the savings calendar is not statistically significant, we cannot reject the null hypothesis that it is equal to the treatment effect of the video in combination with the placebo calendar. Mirroring the result for aspirations, we find that the long-run marginal effects of the video in combination with the savings calendar on agency and pathways — and thus also on hope — are significantly smaller for females. However, in contrast to the results for aspirations, we find that females are overall worse off as a result of the treatment in terms of agency and hope (but not pathways) than had they not received the treatment. This result is intuitive: While the video in combination with the savings calendar does not influence women's pathways (i.e., how to translate their aspirations into reality), their lack of opportunities to follow those pathways negatively affects their sense of agency, thereby lowering their overall hope.

Importantly, despite this, the video in combination with the savings calendar significantly increases the fraction of farmers saving for old age in the long run. In addition, the treatment effect of the video in combination with the savings calendar on the average amount of pension savings is significantly higher among those smallholders who had higher initial levels of total savings. This result mirrors our finding about higher initial levels of total savings predicting higher marginal effects on aspirations and is once again reminiscent of the findings of Gehrke et al. (2023) regarding educational outcomes.

Our study contributes to the growing literature on how psychological and behavioral interventions can promote long-term savings among the rural poor, and more specifically the role of aspiration-raising interventions. While existing papers focus primarily on the effects of aspiration-enhancing interventions on business and educational outcomes (Beaman et al., 2012; Bernard et al., 2019, 2023), to the best of our knowledge, ours is the first to measure their effects on pension savings in a development context. Hence, we contribute to the topical research and policy discourse on old-age welfare of the poor in developing countries.

These findings, while specific to our sample of Ghanaian cocoa farmers, resonate with a broader body of evidence on the power of aspirations in shaping economic outcomes. To further contextualize our results within this existing literature, we first turn to several studies that suggest that exposure to relatable role models can enhance aspirations and downstream outcomes in low- and middle-income settings. Lybbert and Wydick (2017) showed that a short documentary on successful women entrepreneurs, combined with a goal-setting magnet and a workshop on aspirations, agency, and pathways, raised aspirations and modestly increased agency among female community bank members in Mexico after a month. Rojas Valdes et al. (2022) developed a 25-minute video on successful borrowers, supplemented by periodic follow-up messages, and observed positive short-term (1 month) and long-term (1 year) effects on aspirations, agency, and pathways. Cecchi et al. (2022) implemented a similar intervention in Bolivia and found increased production aspirations among dairy farmers. Beaman et al. (2012) found that exposure to female leadership positions in village councils in India, created via a gender quota system, substantially increased educational aspirations and outcomes of young women. Bernard et al. (2019) invited Ethiopian households to watch a one-hour documentary about local individuals who improved their lives, resulting in higher educational aspirations and investments. Riley (2024) showed a 2-hour film about a female role model who succeeded in becoming a chess master through hard work and perseverance to students in Uganda and found improved exam results among viewers compared to a group which watched a placebo film.

Complementary studies start from the premise that aspirations are socially formed. Garcia et al. (2020) found that a microfinance group lending program in Sierra Leone increased life and future economic aspirations. Chiapa et al. (2012) showed that an anti-poverty program in Mexico enhanced parents' educational aspirations for their children. Macours and Vakis (2014) found that interaction with local leaders strengthened poor households' investment response in the context of a cash transfer program in Nicaragua.

The rest of the paper is structured as follows. Section 2 details the study setup and timeline. Section 3 outlines the methodology and Section 4 describes the data. Section 5 presents the empirical results, and Section 6 discusses and concludes.

2 Study setup

2.1 Study setting and partner organization

We focus on smallholder cocoa farmers in Ghana, as a number of factors make this setting ideal for our research. Smallholder farming is of key social and economic importance in the developing world (Fan and Rue, 2020), making smallholder farmers' economic well-being essential for achieving the United Nations' Sustainable Development Goals (SDGs), as smallholder farms often serve as the primary source of income for rural households. In fact, smallholder agriculture is the backbone of rural livelihoods in SSA, where such farms account for the majority of producers and supply a substantial share of food (Lowder et al., 2016), while employing around 60% of the region's population (Sakho-Jimbira and Hathie, 2020).

In Ghana, in particular, agriculture accounts for roughly one-fifth of GDP and around one-third of total employment, while cocoa is the country's most important cash crop and supports the livelihoods of about 800,000 cocoa farmers (as well as their family members, farm laborers, and those involved in the downstream supply chain).[3] However, despite its economic significance, cocoa production is dominated by low-income smallholders who face persistent exposure to income volatility, food insecurity, and poverty. Structural changes in rural Ghana have further heightened old-age vulnerability. Out-migration of younger household members — driven by, e.g., urbanization, rising educational aspirations and shifting social norms — has reduced the availability of intergenerational support, weakening the traditional family-based safety net (Bezu and Holden, 2014; Amon-Armah et al., 2023). At the same time, formal old-age protection remains limited: Contributory pension schemes cover only a minority of workers, with most rural and informal-sector adults lacking access to retirement benefits. As a result, many smallholder farmers enter old age with little accumulated savings and few reliable sources of support. These conditions underline the policy relevance of interventions aimed at increasing pension savings among rural households and improving their financial security in later life.

To implement our study, we collaborated with People's Pension Trust (PPT), a licensed pen-

[3] Source: World Development Indicators by World Bank (2023); and Ministry of Food and Agriculture (2021).

sion administrator for both low-income formal workers and informal workers in Ghana. When PPT expanded its voluntary contributory pension scheme to rural areas — including the Fanteakwa and Suhum cocoa districts in Ghana's Eastern Region — it provided smallholder farmers with the opportunity to save for old age, but observed very low usage of its pension savings product among registered farmers. Our sample consists of smallholder cocoa farmers in these cocoa districts who had already opened PPT accounts. This makes the sample well-suited to our research question, as these farmers do not face binding external constraints to saving formally for old age, making this a setting where the effects of treatments targeting internal constraints can be clearly identified. At the same time, since the challenges confronting Ghanaian cocoa farmers in saving for old age are broadly similar to those faced by other smallholders in SSA, the setting also offers useful insights that are likely to extend beyond this particular context.

2.2 Interventions and experimental design

Participants were randomly assigned to one of three groups: a control group which received a placebo calendar; a treatment group that received a video plus placebo calendar (*Vid. + Plac. Cal.*) treatment; and a treatment group that received a video plus savings calendar (*Vid. + Sav. Cal.*) treatment.

Control group. The control group received a regular calendar as placebo, which was an annual wall calendar with a decorative picture of a rose (see Appendix Figure A.1 for the design). Control group participants were given this calendar to treat them on a par with treatment group participants, to prevent frictions that might have affected outcome measures.

Treatment group: *Vid. + Plac. Cal.* Farmers in this group were shown a video and were given the same placebo calendar as the control group (to prevent frictions, but also to rule out the interpretation that the savings calendar in the *Vid. + Sav. Cal.* treatment group simply served as a reminder of the video intervention). The video focused on positive testimonies from farmers who saved toward old age and realized their aspirations. The content emphasized how regular, small savings can improve old-age well-being for cocoa farmers (see screenshot from the video in Appendix Figure A.2), without mentioning how to save (i.e., we did not promote any specific ways to save for old age or mention any specific savings products). Those featured in the video (i.e., the actors) were actual cocoa farmers, from similar communities as the ones of the farmers in our sample, though they belonged to a different cocoa district. The video had a running time of 5 minutes 21 seconds, out of which men shared their experiences/opinions for 2 mins 41 seconds and women for 1 min 15 seconds, and of which 22 seconds were devoted to a young man asking a question. Farmers were semi-scripted. The voices were recorded in Twi, which is the main language spoken in the area, and the video was accompanied by English subtitles. The video was stored on the tablet of each enumerator and was shown to each participant individually and privately to avoid external influences such as community effects. Participants who had watched the video and completed the midline were asked to immediately leave the location in order to prevent discussions of the video with other participants (see also Section 2.3 for the full timeline of the intervention).

Treatment group: *Vid. + Sav. Cal.* Individuals in this group were shown the video described above and received a savings calendar. This calendar was also an annual wall calendar, like the placebo calendar, but differed from the latter in several important ways (see Appendix Figure A.1 for the design).

Instead of the picture of a rose as in the placebo calendar, the savings calendar depicted a young cocoa tree, relating it to the principal income-generating activity for our participants, and displayed other prominent motifs suggestive of progressive savings. The design also featured two empty white rectangular spaces at the base of the tree where the owner of the calendar could indicate (i) a concrete, non-monetary, long-term savings goal for old age (e.g., owning a house), and (ii) a monthly financial savings target toward achieving this goal. Participants in this treatment group were asked, after watching the video, to pencil (i) and (ii) into the two blank spaces provided in the savings calendar. These two self-inscriptions were meant to increase the salience of the need to save for old age and of farmers' specific savings goals for old age; hence they were encouraged to hang these calendars where they would see them every day.

2.3 Timeline and data collection procedure

The timeline of our field activities is depicted in Figure 1. The interventions and baseline data collection were carried out from 19 September 2019 to 1 November 2019. Farmers were asked to report to a community center, a community school, or a church building and were then interviewed individually, to ensure privacy, by trained enumerators using a structured questionnaire. To limit enumerator influence on farmers' responses, questions regarding aspirations and hope (i.e., agency and pathways) were pre-recorded and played to each participant. Those in either of the two treatment groups were then shown the video on a tablet. All participants were asked to

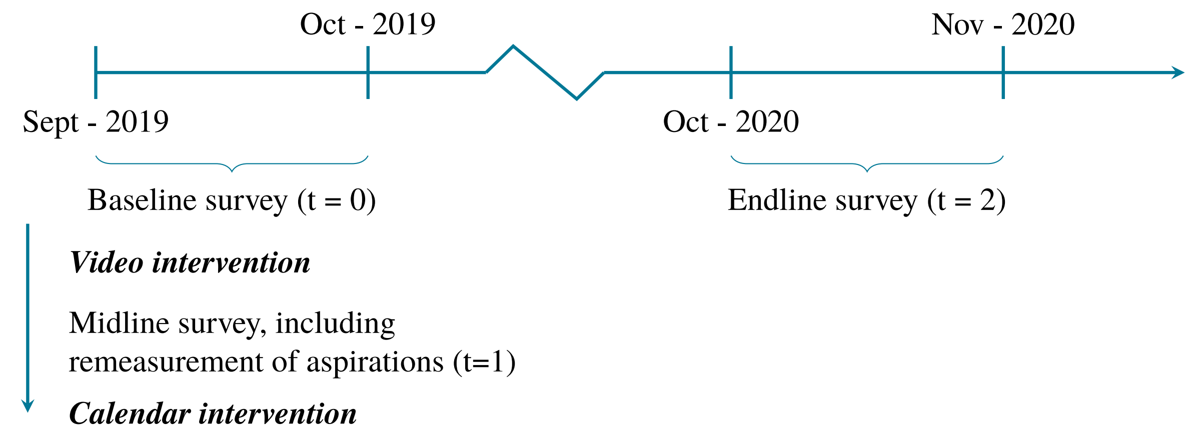

Figure 1: Timeline of the study

respond to the aspirations statement for a second time (hereafter, the midline survey, reflecting its timing). The endline took place between 13 October 2020 and 23 November 2020.[4] Note that data collection was done by the Social Science and Statistics Unit of the Cocoa Research Institute of Ghana (CRIG), the latter being a subsidiary of the Ghana Cocoa Board (COCOBOD). Thus, our enumerators were not affiliated with PPT.

3 Methodology

Our general approach to study treatment effects is to use analysis of covariance (ANCOVA).[5] For *Aspirations*, which was measured thrice (at baseline, at midline and at endline; see Section 2.3), we estimate the immediate average treatment effect using *Aspirations* measured at midline, as well as the long-term average treatment effect using *Aspirations* measured at endline. For the *Hope*, *Agency* and *Pathways* indices, as well as the savings outcomes, we assess long-term average treatment effects. In addition, for all psychological and savings outcomes, we assess heterogeneity in long-term treatment effects using selected baseline characteristics, as outlined in the introduction.

In all cases, we regress our outcome variables on their baseline values and the treatment dummies, while controlling for variables selected on the basis of our balance tests and attrition analysis.[6]

[4] The endline survey was initially planned to be administered six months after the baseline (i.e., March to April 2020) but was disrupted by nationwide restrictions in response to the Covid-19 pandemic. Thus, we conducted the endline survey from October to November 2020, once Covid-19 restrictions were relaxed sufficiently.

[5] We use ANCOVA for all outcome variables except for PPT savings, for which we do not have baseline values. PPT savings are analyzed using simple post-treatment estimates, for which the equations estimated are identical to (1)–(3), except that they do not include the baseline value of the dependent variable as a regressor.

[6] We also select controls using a double LASSO methodology as a robustness check when estimating treatment

3.1 Short-term effect of video intervention

To estimate the short-term average treatment effect of the video on *Aspirations*, we use the following regression:

$$Y_{i,1} = \alpha + \beta_1 Video_i + \beta_2 Y_{i,0} + \gamma X_{i,0} + \varepsilon_i, \quad (1)$$

where subscript i represents the i-th individual and 0 represents a baseline measurement; 1 represents a midline measurement; Y represents *Aspirations*; the dummy variable *Video* is an indicator for having watched the aspirations video, i.e., for belonging to either of the two treatment groups, and X are the control variables.

3.2 Long-term effect of interventions on psychological and savings outcomes

Proceeding in an analogous manner, we assess the long-term effect of the interventions on the psychological and savings outcomes by estimating:

$$Y_{i,2} = \alpha + \beta_1 (Video + Plac.\ Cal.)_i + \beta_2 (Video + Sav.\ Cal.)_i + \beta_3 Y_{i,0} + \gamma X_{i,0} + \varepsilon_i, \quad (2)$$

where subscript 2 represents an endline measurement. All other notation remains unchanged.

To estimate the marginal effect of the savings calendar in the long term, we define a dummy variable, $AnyVid_i$, that equals one if individual i watched the video, regardless of treatment arm. Proceeding as before, we estimate the following model, where the coefficient on $AnyVid_i$ is the average marginal effect of having watched the video as part of *either* treatment group and the coefficient on $(Video + Sav.\ Cal.)_i$ is the marginal effect of the savings calendar over and above that effect, both in the long run:

$$Y_{i,2} = \alpha + \beta_1 AnyVid_i + \beta_2 (Video + Sav.\ Cal.)_i + \beta_3 Y_{i,0} + \gamma X_{i,0} + \varepsilon_i. \quad (3)$$

3.3 Long-term heterogeneous treatment effects

Following the arguments presented in the introduction, we do not expect treatment effects to be the same across gender and initial level of savings. Hence, we conduct heterogeneity analyses

effects on the psychological outcome variables. This does not affect the results (see Appendix F).

of treatment effects with respect to these key variables as outlined below.

Specifically, we estimate the following equation, considering interactions between treatments and the selected demographic or socio-economic variable:

$$Y_{i,2} = \alpha + \beta_1 \, (Video + Plac.\ Cal.)_i \times D_{i,0} + \beta_2 \, (Video + Sav.\ Cal.)_i \times D_{i,0} + \beta_3 Y_{i,0} + \gamma Z_{i,0} + e_i, \tag{4}$$

where D is the baseline (0) value of the variable of interest, i.e., gender or total savings, at the individual level (i), and $Z_{i,0}$ is the union of $D_{i,0}$ and our earlier set of baseline controls. All other notation remains the same.

4 Data

4.1 Sampling frame, randomization and determination of planned sample

The sampling frame for our study consisted of all cocoa farmers belonging to the Fanteakwa and Suhum cocoa farming cooperatives in the Eastern Region of Ghana who were registered with PPT, i.e., who had opened pension savings accounts up until August 2019. These farmers belonged to 56 communities. As a maximum of 22 individuals per community were (randomly) chosen, the number of eligible individuals per community varied widely, based on the size of a community, ranging from 1 to 22. This led to a planned sample size of 956 farmers. Stratification was then done at the community level, with eligible individuals being randomly assigned to the control and two treatment groups.

4.2 Communication prior to interventions

Each cooperative operated with an overall chairperson and executive members overseeing local chairpersons at the community level. To maximize the size of our realized sample from the sampling frame described above, we contacted the overall chairperson of each of the two cooperatives, and in subsequent visits to their offices briefed them and the executive members about the survey we wanted to undertake and the objectives of the project in general terms. The next step was to schedule the surveys and interventions in the field. To do so, we contacted the local community chairpersons by phone, when possible, about two weeks before the start of the surveys (by which time they had been made aware of our project by the overall chairperson of their cooperative). In some cases, where phone contact was unsuccessful, we had to travel to the community to make arrangements. In the majority of cases, to ensure maximum participation, the surveys and interventions were scheduled on either the local cooperative meeting day or the local "taboo day".[7] For most communities, these two days coincided. We asked the local chairpersons to inform, on our behalf, the farmers we had included in our sample that there would be a survey (executed by a CRIG representative, CRIG being an organization with which the farmers had strong prior ties). Contacting the chairpersons two weeks in advance provided them with ample opportunity to inform those selected, and also to remind them of the survey during the following week's meeting. We called or visited the local chairpersons again within a week after the first call or visit, and visited them as an additional reminder when their communities were next in line to be surveyed.

On the day of the meeting and before the survey took place, the project objective was explained to each farmer in general terms. We also gave them an estimate of the time it would take to complete the survey and informed them that they would receive a cutlass (a commonly used tool in cocoa farming) as an in-kind show-up fee. The farmers were then asked to sign a consent form if they wished to participate in the study.

4.3 Realized sample

A total of 723 and 620 cocoa farmers out of the planned sample were interviewed for the baseline and endline, respectively (see last column of Table 1). The first row of Table 1 shows the group assignments of the 723 cocoa farmers who could be interviewed for the baseline: 235 farmers were assigned to the control group, 240 farmers to the Vid. + Plac. Cal. treatment group, and 248 to the Vid. + Sav. Cal. treatment group. For the endline, 197 of those in the control group, 208 of those in the Vid. + Plac. Cal. treatment group, and 215 of those in the Vid. + Sav. Cal. treatment group could be successfully re-interviewed.

4.4 Outcome variables

Psychological outcome variables: Aspirations and Hope. Our psychological outcome variables are *Aspirations* regarding financial well-being in old age and three hope-related indices, i.e., *Hope*, and its two sub-indices *Agency* and *Pathways*. To construct these four variables,

[7] "Taboo days" are days on which farmers do not farm, to provide rest for both the land and themselves.

Table 1: Realized sample across survey rounds and treatment groups

Survey Round	Control	Vid. + Plac. Cal.	Vid. + Sav. Cal.	Sum
Baseline	235	240	248	723
Endline	197	208	215	620

Notes: The first and second rows show the group assignment of those cocoa farmers interviewed for the baseline and endline surveys, respectively.

Table 2: Statements used to construct the psychological outcome variables

Outcome	**Statement**
Aspirations	1: It is important to set financial goals when planning for old age / the future.
Hope	
Agency	1: "You've been pretty successful in life". 2: "You energetically pursue your goals". 3: "Your past experiences have prepared you well for your future". 4: "You meet the goals that you set for yourself".
Pathways	1: "You can think of many ways to get out of a jam". 2: "There are lots of ways around any problem". 3: "You can think of many ways to get the things in life that are important to you". 4: "Even when others get discouraged, you know you can find a way to solve the problem".

Notes: The table lists the statements used to construct our main psychological outcome variables: *Aspirations* and *Hope* (i.e., *Agency* and *Pathways*, see text for further details). The Aspirations statement was adapted from Lybbert and Wydick (2018) to the context of old age. The Hope index was constructed using four statements measuring Agency and four statements measuring Pathways, retained in their original wording from the Adult Hope Scale (see Snyder, 1994).

we used nine statements (see Table 2) based on existing work, one concerning Aspirations, and four each concerning *Agency* and *Pathways*. Specifically, the *Aspirations* statement was adapted from Lybbert and Wydick (2018) to the context of old age. The *Hope* index consisted of four statements that measured *Agency* and four statements that measured *Pathways*, which were retained in their original wording from the Adult Hope Scale (see Snyder, 1994).

We asked subjects to indicate their level of agreement with each of the statements on a six-point Likert scale ranging from 0 (strongly disagree) to 5 (strongly agree). The resulting *Aspirations* scores were standardized by subtracting the mean for the control group at baseline and dividing by the standard deviation of the control group at baseline (Kling et al., 2007;

Lybbert and Wydick, 2018), as shown below (equation 5):

$$\frac{e - \bar{e}_{bc}}{\sigma_{\bar{e}_{bc}}}, \tag{5}$$

where e is the Likert scale level of the response, and $\bar{e}_{bc}$ and $\sigma_{\bar{e}_{bc}}$ are respectively the mean and standard deviation of the score for the control group measured at the baseline.

The approach was nearly identical for *Agency*, *Pathways*, and *Hope*, except that the average (ν below) of the responses to the corresponding questions was computed first (i.e., by dividing the sum of the responses to the corresponding questions by the number of corresponding questions), and the mean and standard deviation of ν were then used for standardization, as shown below (see equation 6):

$$\frac{\nu - \bar{\nu}_{bc}}{\sigma_{\bar{\nu}_{bc}}}, \tag{6}$$

where $\nu = \left(\frac{\sum e}{\text{number of questions}}\right)$ is the average of the Likert scale levels of the responses e for all corresponding questions, and $\bar{\nu}_{bc}$ and $\sigma_{\bar{\nu}_{bc}}$ are respectively the mean and standard deviations of ν for the control group measured at the baseline.

Outcome variables related to pension savings. In addition to the psychological outcome variables described above, we considered the impact of the interventions on relevant savings outcomes, specifically, the fraction of farmers who report saving for old age (*Saving for old age*) as well as the amount of savings farmers store in their PPT accounts, measured in Ghana Cedis (*PPT pension savings*).

4.5 Explanatory variables

Besides the outcome variables, we collected additional variables via the baseline and endline surveys. These variables include demographics, socio-economic status as well as shocks faced by households (see Table 3).

4.6 Descriptive statistics, balance and attrition

Table 3 provides summary statistics of our sample at baseline. Respondents have high levels of aspirations (a non-standardized average score of 3.58 out of a maximum of 5) and hope (a non-standardized average score of 4.05 out of a maximum of 5) to begin with. Consistent with the general demographics of cocoa farmers in Ghana, the study respondents are predominantly

Table 3: Descriptive statistics

Category	Variables	N	Mean	SD	Min	Max
Psychological outcome variables	Aspirations (Bsl.)	723	0.040	0.951	-7.659	0.666
	Agency (Bsl.)	723	0.090	1.018	-3.521	1.595
	Pathways (Bsl.)	723	0.028	1.076	-5.981	1.337
Old age salience	Salience of old age	723	0.678	0.468	0	1
Demographics	Female	723	0.196	0.398	0	1
	Age	721	55.854	12.547	24	97
	Household size	723	5.660	2.460	1	23
	Migration status	722	0.309	0.462	0	1
	Speaks Twi	723	0.819	0.385	0	1
	Education	705	9.275	2.833	0	20
	Financial literacy	723	0.642	0.480	0	1
Income and savings	Income	662	11.261	71.293	0	1,806.9
	Total savings (Bsl.)	423	1.310	2.833	0	31.2
	Saving for old age	723	0.570	0.495	0	1
Other assets	Total farm size	717	9.333	19.376	0	252
	Farm equipment	723	0.845	0.662	0	2
	House size	723	4.113	2.407	1	15
	Elec. and electr. assets	723	3.195	1.572	0	8
	Vehicles	723	0.451	0.646	0	3
Social standing	Leadership status	723	0.159	0.366	0	1
	Social capital	723	2.183	1.184	0	4
Shocks	Financial difficulty	723	0.172	0.377	0	1
	Natural/biological disaster	723	0.351	0.478	0	1

Notes: The table presents descriptive statistics. Salience of old age = 1 if the respondent often thinks about old age; Female = female respondent; Age = respondent's age; Household size = number of people in the respondent's household (defined as individuals who live together in the same dwelling, share the same housekeeping arrangements, and are catered for as one unit under one recognized head); Migration status = 1 for first-generation migrants; Speaks Twi = 1 if the respondent speaks Twi; Education is years of formal schooling; Financial literacy = 1 if the respondent answers at least three out of four questions correctly (one math, one inflation, and two interest rate questions); Income is calculated as reported annual cocoa yield multiplied by GH¢475 per 64 kg bag (the government's announced producer price of cocoa for the 2018/2019 cocoa season that ended on 30 September 2019); Total savings (Bsl.) is the total savings of the respondent at baseline measured in GH¢; Saving for old age = 1 if the respondent saves for old age; Total farm size is the total farm acreage owned by the respondent; Farm equipment is the number of categories of farm equipment owned by the respondent, mainly including knapsack sprayers and mist blowers; House size is the number of rooms in the respondent's house. Elec. and electr. assets is the number of categories of electric and electronic assets owned by the respondent (e.g., fridge, freezer, radio, satellite dish, television, cellphone); Vehicles is the number of categories of transportation assets owned by the respondent (e.g., bicycle, tricycle, motorbike); Leadership status = 1 if the respondent holds any leadership role in the community, farmer group, or association; Social capital is the number of important people within the community that the respondent is connected to (e.g., the chief, spiritual leader, farmer group leader); Financial difficulty = 1 if the respondent sold assets due to debt distress in the 12 months prior to the intervention; Natural/biological disaster = 1 if the respondent experienced any natural disaster or their farm(s) experienced an unusually high level of pests or disease in the 12 months prior to the intervention.

male (80.4%) with only 19.6% being female. Their mean age is 55.9 years, with an average household size of almost 6 and an average of 9.3 years of formal education. Respondents have an average of 9.3 acres of self-owned cocoa farm.

It is important to note that the realized baseline sample (see Table 1) is smaller than the planned sample that was randomized into the control and treatment groups. Nevertheless, the realized baseline sample remains well-balanced: The balancing tests, as shown in Appendix Table B.4, indicate that the randomization was largely successful, with only a few variables exhibiting significant differences in means across the three groups. In particular, out of the 69 comparisons displayed in the balance table, three are significant at the 5% level (including one at the 1% level). This is close to the 3.45 and 0.69 differences expected by chance at the 5% and 1% levels, respectively.

Yet, to be cautious, we present our results both with and without controlling for all variables that exhibit significant differences (including at the 10% level).[8] For reasons of space, the specifications without controls are reported in Appendix C, Part A, with the exception of the short-run treatment effect on aspirations, which is also shown in the main text.

In addition, as mentioned earlier, some level of attrition at endline was observed:[9] 723 farmers participated in the baseline survey, 620 farmers in the endline (see Appendix Table B.1). To test whether attrition was non-random, we ran two regressions as follows:

$$Attendance_{i,2} = \alpha + \beta_1 \,(\text{Video} + \text{Plac. Cal.})_i + \beta_2 \,(\text{Video} + \text{Sav. Cal.})_i + e_i, \tag{7}$$

and

$$Attendance_{i,2} = \alpha + \beta_1 \,(\text{Video} + \text{Plac. Cal.})_i + \beta_2 \,(\text{Video} + \text{Sav. Cal.})_i + \gamma W_{i,0} + e_i, \tag{8}$$

where *Attendance* is an indicator variable for whether a respondent was interviewed at endline (1 for Yes); W is a vector of outcome and explanatory variables, and the remaining notation is as before.

The attrition tests (see Appendix Tables B.2 and B.3) indicate just one variable (*Salience of old age*) to be a negative predictor of attrition, which we control for in all the subsequent analyses where we also control for variables that are unbalanced at baseline.

[8] These variables are *Female*, *Age*, *Migration status*, *Farm equipment*, *Social capital*, *Financial difficulty* and *Natural/biological disaster*.

[9] Note that there was no attrition at midline.

Table 4: Short-term treatment effect of video intervention on aspirations

VARIABLES	(1) Aspirations (Mdl.)	(2) Aspirations (Mdl.)
Video	0.149**	0.120*
	(0.073)	(0.070)
Aspirations (Bsl.)	0.513***	0.518***
	(0.074)	(0.075)
Constant	0.044	0.382*
	(0.061)	(0.195)
Baseline controls	No	Yes
Observations	618	618
R-squared	0.262	0.282

Notes: The table presents the results of an ANCOVA specification to assess the video treatment effect on aspirations. Columns 1 and 2 present estimates without and with control variables, respectively. Baseline controls in Column 2 consist of the variables that were unbalanced at baseline (Female; Age; Migration status; Farm equipment; Social capital; Financial difficulty; Natural/biological disaster) and the attrition predictor (Salience of old age). See Table 3 for variable definitions. Full regression showing coefficients of the baseline controls is presented in Appendix Table C.5. Huber-White robust standard errors are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

5 Empirical results

5.1 Short-term effect of video intervention on aspirations

Recall that we recorded participants' aspirations at baseline (*Aspirations (Bsl.)*). We then recorded aspirations a second time: for the control group immediately after the baseline, and for the treatment groups, after watching the video (*Aspirations (Mdl.)* in both cases). We estimate the short-term effect of the video intervention on aspirations using this second measure. Column 1 of Table 4 shows that, without controlling for baseline variables, the video intervention increases the aspiration score by 0.15 standard deviations (significant at 5%). When adding controls, the estimated effect decreases to 0.12 standard deviations and remains significant at the 10% level.[10]

These results are in line with previous studies that report positive short- to medium-term impacts of similar interventions (e.g., Bernard et al., 2023; Ross et al., 2021; Bhan, 2021; Orkin et al., 2023; Lybbert and Wydick, 2018; Rojas Valdes et al., 2022) intended to enhance psychological attributes such as aspirations. For instance, Rojas Valdes et al. (2022) found a significant positive short-term (one month after the intervention) effect of a video intervention on the aspi-

[10]For reasons of space, we relegate the full regression results (including control coefficients) to Appendix C, Part B (Appendix Table C.5).

rations of female entrepreneurs.

5.2 Long-term treatment effects on psychological outcome variables

For reasons of space, we present the estimates of all long-term treatment effects on psychological outcome variables with controls in the main text, while relegating the results without controls to Appendix C, Part A, and the full regression results (including control coefficients) to Appendix C, Part B.

Aspirations. Does the significant immediate increase in measured aspirations uncovered in Section 5.1 persist in the longer run? Recall from Section 2.3 that, after recording the immediate impact of the video intervention, we implemented the savings calendar intervention, which was designed to sustain the increases in aspirations (and hope) over the longer term by increasing the salience of the need to save for old age and of farmers' specific savings goals for old age. The video plus placebo calendar treatment group and the control group both received a placebo calendar. Table 5 presents ANCOVA estimates of the effects of the two treatments, the video with the placebo calendar (*Vid. + Plac. Cal.*) and the video with the savings calendar (*Vid. + Sav. Cal.*), on long-term aspirations (measured one year after the interventions).

The results reveal no statistically significant long-term average treatment effect on subjects' aspirations. The significant short-term effect on aspirations resulting from watching the video, which was observed in Table 4, appears to have dissipated in the longer run. These findings align with the broader literature on aspiration-raising interventions. Bernard et al. (2023) similarly found that exposure to local role models in Ethiopia increased aspirations in the short term, but that these effects faded over time. Rojas Valdes et al. (2022) also observed that aspiration gains dissipated after one year. Comparable interventions by Cecchi et al. (2022) and Beaman et al. (2012) likewise found that exposure to relatable role models can enhance aspirations and goal-directed behavior, though sustaining these effects over time remains a challenge.

Hope and its components, Agency and Pathways. Unlike the results for the average long-term impact of our treatments on aspirations, we find a statistically significant positive long-run average treatment effect of the *Vid. + Plac. Cal.* treatment on subjects' hope: for participants who received the *Vid. + Plac. Cal.* treatment, *Hope* increased by 0.142 standard deviations (see Column 2 of Table 5). This effect is driven by an increase in perceived agency, with the *Vid. + Plac. Cal.* treatment increasing *Agency* by 0.147 standard deviations (see Column 3 of Table

Table 5: Long-run treatment effects on psychological outcome variables (with controls)

VARIABLES	(1) Aspirations	(2) Hope	(3) Agency	(4) Pathways
Video + Plac. Cal.	0.017	0.142**	0.147**	0.064
	(0.082)	(0.072)	(0.075)	(0.068)
Video + Sav. Cal.	0.071	0.058	0.042	0.057
	(0.084)	(0.068)	(0.073)	(0.070)
Aspirations (Bsl.)	0.012			
	(0.039)			
Hope (Bsl.)		0.132***		
		(0.030)		
Agency (Bsl.)			0.147***	
			(0.030)	
Pathways (Bsl.)				0.033
				(0.024)
Constant	-0.844***	-0.335**	-0.181	-0.407***
	(0.196)	(0.164)	(0.177)	(0.149)
Baseline controls	Yes	Yes	Yes	Yes
Observations	618	618	618	618
R-squared	0.035	0.085	0.088	0.048
Test of differences in treatment coefficients	F(1, 606) = 0.41 Prob>F=0.520	F(1,606)=1.57 Prob>F=0.211	F(1,606)=2.46 Prob>F=0.117	F(1,606)=0.01 Prob>F=0.925

Notes: The table presents the results of an ANCOVA specification to assess treatment effects on aspirations, hope, agency, and pathways. The treatment variables are video plus placebo calendar (Vid. + Plac. Cal.) and video plus savings calendar (Vid. + Sav. Cal.). Baseline controls consist of the variables that were unbalanced at baseline (Female; Age; Migration status; Farm equipment; Social capital; Financial difficulty; Natural/biological disaster) and the attrition predictor (Salience of old age). See Table 3 for variable definitions. Full regressions showing coefficients of the baseline controls are presented in Appendix Table C.5. Huber-White robust standard errors are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

5). Although the corresponding average treatment effects for the *Vid. + Sav. Cal.* treatment are not statistically significant, we cannot reject the null hypothesis that the treatment effects of the two treatments are equal.

Since these findings suggest that the video, rather than the calendar intervention, may be the primary driver of the observed increases in *Hope* and *Agency*, we estimate the marginal effect of having watched the video as part of either treatment group (*AnyVid*) and the marginal effect of the savings calendar over and above that effect, as explained in Section 3.2. We find that the coefficient on *AnyVid* is positive and statistically significant, while the coefficient on *Vid. + Sav. Cal.* is not (see Appendix Table D.1). This provides further evidence that the video intervention positively enhanced subjects' *Hope* (via *Agency*) in the long term.

At the same time, we do not find any statistically significant long-run average treatment effects on *Pathways*. This is consistent with some previous research but contrasts with other studies. For instance, Lybbert and Wydick (2018) report no treatment effect on pathways one month after their intervention, despite it including training on envisioning pathways out of poverty. Appadurai (2007) suggests that pathways may be particularly rigid for the poor. In contrast, Rojas Valdes et al. (2022) find a significant positive effect on pathways 12 months after their video intervention. In our case, the video intervention focused on positive testimonies from cocoa farmers who saved toward old age and realized their aspirations, emphasizing that regular, small savings can improve well-being in old age, but without offering concrete guidance on how to save or which products to use. Given that neither our video nor our calendar intervention specifically targeted pathways for pension savings, the null results are not surprising.

5.3 Long-run heterogeneous treatment effects on psychological outcome variables

We present the estimates of all long-run heterogeneous treatment effects on psychological outcome variables with controls in the main text and relegate the results without controls to Appendix C, Part A, and the full regression results (including control coefficients) to Appendix C, Part B.

Gender. The introduction outlined our choice of variables for the heterogeneity analysis (gender and initial level of savings). Table 6 (Column 1) presents ANCOVA estimates showing a statistically significant difference in the effect of the *Vid. + Sav. Cal.* treatment on aspirations for women. Specifically, the treatment effect on aspirations is lower for women than for men by 0.461 standard deviations.[11] Mirroring this result, we find that the long-run effects of the *Vid. + Sav. Cal.* treatment on *Agency* and *Pathways* – and thus also on *Hope* – are statistically significantly smaller for women as well (see Columns 2-4 in Table 6). More precisely, being female is associated with reductions in the *Vid. + Sav. Cal.* treatment effects on *Hope*, *Agency* and *Pathways* by 0.451, 0.422, and 0.347 standard deviations, respectively. However, in contrast to the result for aspirations, we find that women are also overall (statistically significantly) worse off as a result of the treatment in terms of *Hope* and *Agency* (but not *Pathways*) compared to not receiving the treatment, as shown by the Lincom estimates in Table 6.

This result is intuitive, as it may reflect that women in our sample face greater obstacles in turning aspirations into reality compared to male participants: Recall that the savings calendar

[11] Note that the Lincom estimates in Table 6 show that women are overall not statistically significantly worse off as a result of the treatment (as compared to not receiving the treatment).

Table 6: Long-run heterogeneous treatment effects by gender on psychological outcome variables (with controls)

VARIABLES	(1) Aspirations	(2) Hope	(3) Agency	(4) Pathways
Video + Plac. Cal.	0.019	0.146*	0.166**	0.050
	(0.089)	(0.080)	(0.084)	(0.075)
Video + Sav. Cal.	0.154*	0.139*	0.116	0.122
	(0.090)	(0.075)	(0.081)	(0.076)
Female	0.224	0.318**	0.346**	0.175
	(0.192)	(0.148)	(0.149)	(0.149)
Female × (Video + Plac. Cal.)	-0.111	-0.119	-0.181	-0.010
	(0.229)	(0.188)	(0.181)	(0.190)
Female × (Video + Sav. Cal.)	-0.461*	-0.451**	-0.422**	-0.347*
	(0.242)	(0.187)	(0.184)	(0.198)
Aspirations (Bsl.)	0.007			
	(0.039)			
Hope (Bsl.)		0.125***		
		(0.030)		
Agency (Bsl.)			0.142***	
			(0.030)	
Pathways (Bsl.)				0.029
				(0.024)
Constant	-0.920***	-0.409**	-0.256	-0.458***
	(0.194)	(0.168)	(0.180)	(0.155)
Baseline controls	Yes	Yes	Yes	Yes
Observations	618	618	618	618
R-squared	0.043	0.095	0.096	0.057
Lincom: (Video + Sav. Cal.) +	-0.307	-0.311*	-0.306*	-0.225
(Female × (Video + Sav. Cal.))	(0.227)	(0.170)	(0.165)	(0.181)

Notes: The table presents the results of an ANCOVA specification to assess the heterogeneous treatment effect by gender (Female) on aspirations, hope, agency and pathways. The treatment variables are video plus placebo calendar (Vid. + Plac. Cal.) and video plus savings calendar (Vid. + Sav. Cal.). Baseline controls consist of the variables that were unbalanced at baseline (Age; Migration status; Farm equipment; Social capital; Financial difficulty; Natural/biological disaster), excluding Female, and the attrition predictor (Salience of old age). See Table 3 for variable definitions. Full regressions showing coefficients of the baseline controls are presented in Appendix Table C.6. Huber-White robust standard errors are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

intervention was designed to increase the salience of the need to save for old age and of specific savings goals by requiring farmers to write these goals into the empty fields of the calendar. While participants in the *Vid. + Plac. Cal.* treatment, who did not write down their goals, could more easily adjust them to changing external circumstances, the written goals in the *Vid. + Sav. Cal.* treatment may have been perceived as a commitment in the presence of adverse external shocks. The inability to meet such commitments likely frustrated participants' sense of agency, thereby reducing their overall hope. This pattern is consistent with the broader gender disparities discussed above: women's limited access to resources and decision-making power may make it harder for them to act on their aspirations, especially when external shocks compound these constraints.

Initial total savings. Table 7 presents ANCOVA results of heterogeneous treatment effects by total savings at baseline on aspirations. We find an enhancing effect of both the *Vid. + Plac. Cal.* and the *Vid. + Sav. Cal.* treatments on aspirations for those who saved more at baseline. Specifically, every 1000 Ghana cedis (GH¢) increase in total savings at baseline is associated with the treatment effects on aspirations being higher by 0.086 standard deviations (see Column 1 of Table 7). This result echoes related literature (see Gehrke et al., 2023), which finds that treatment effects on educational aspirations are larger for those who received better grades at baseline; analogously, our effects are stronger for those who already saved more at baseline. As for hope and its components, we do not find any statistically significant heterogeneous effects with respect to initial savings.

5.4 Long-run treatment effects on savings outcomes, including heterogeneous treatment effects

Once again, we present the estimates of all long-term treatment effects with controls in the main text and relegate the results without controls to Appendix C, Part A, and the full regression results (including control coefficients) to Appendix C, Part B.

Saving for old age — extensive margin. The ANCOVA results presented in Table 8 (Columns 1-3) show the effects of the *Vid. + Plac. Cal.* and *Vid. + Sav. Cal.* treatments on the probability that subjects save for old age. The findings reveal that subjects who received the *Vid. + Sav. Cal.* treatment were 17.2 percentage points more likely to save for old age compared to the control group (Column 1), suggesting that the combination of the video and the savings calendar positively affected the likelihood of saving for retirement in the long run. Column 2 examines heterogeneity by gender and shows no statistically significant differences, indicating that the effectiveness of the intervention in promoting saving for old age did not statistically significantly differ between male and female participants. Column 3 examines heterogeneity by total baseline savings and likewise reveals no statistically significant effects, suggesting that the impact of the interventions did not vary significantly with participants' initial savings levels.

Table 7: Long-run heterogeneous treatment effects by initial level of total savings on psychological outcome variables (with controls)

VARIABLES	(1) Aspirations	(2) Hope	(3) Agency	(4) Pathways
Video + Plac. Cal.	-0.048	0.168*	0.166	0.071
	(0.119)	(0.096)	(0.101)	(0.086)
Video + Sav. Cal.	-0.085	-0.007	-0.047	0.041
	(0.121)	(0.100)	(0.104)	(0.093)
Total savings (Bsl.)	-0.023***	0.003	0.004	-0.006
	(0.008)	(0.009)	(0.010)	(0.006)
Total savings (Bsl.) × (Video + Plac. Cal.)	0.072**	0.013	0.030	-0.006
	(0.032)	(0.021)	(0.022)	(0.018)
Total savings (Bsl.) × (Video + Sav. Cal.)	0.086**	0.049	0.047	0.033
	(0.035)	(0.036)	(0.040)	(0.015)
Aspirations (Bsl.)	-0.002			
	(0.043)			
Hope (Bsl.)		0.136***		
		(0.038)		
Agency (Bsl.)			0.164***	
			(0.038)	
Pathways (Bsl.)				0.017
				(0.028)
Constant		-0.399*	-0.098	-0.606***
		(0.221)	(0.238)	(0.192)
Baseline controls	Yes	Yes	Yes	Yes
Observations	367	367	367	367
R-squared	0.036	0.092	0.127	0.040
Lincom: (Video + Sav. Cal.) +	0.001	0.043	0.000	0.074
(Total savings (Bsl.) × (Video + Sav. Cal.))	(0.111)	(0.088)	(0.092)	(0.083)

Notes: The table presents the results of an ANCOVA specification to assess the heterogeneous treatment effect by baseline total savings (Total savings (Bsl.)) on aspirations, hope, agency and pathways. The treatment variables are video plus placebo calendar (Vid. + Plac. Cal.) and video plus savings calendar (Vid. + Sav. Cal.). Baseline controls consist of the variables that were unbalanced at baseline (Female; Age; Migration status; Farm equipment; Social capital; Financial difficulty; Natural/biological disaster) and the attrition predictor (Salience of old age). See Table 3 for variable definitions. Full regressions showing coefficients of the baseline controls are presented in Appendix Table C.7. Huber-White robust standard errors are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Pension savings — intensive margin. Our analysis reveals a marginally significant and positive effect of the *Vid. + Sav. Cal.* treatment on pension saving with PPT. Farmers who received this treatment increased their pension savings by approximately GH¢ 39 (see Column 4 of Table 8). To contextualize this amount, a day's wage for a farm laborer was around GH¢ 50 at the time of the intervention.

Due to non-responses, the number of observations used for these estimates is substantially lower. To assess whether non-responses are systematically related to the outcome variable, we

Table 8: Long-run treatment effects on savings outcomes (with controls), including heterogeneous treatment effects by gender and initial level of total savings

	Saving for old age (Endl.)			PPT pension savings		
VARIABLES	(1)	(2)	(3)	(4)	(5)	(6)
Video + Plac. Cal.	0.004	-0.017	0.110	28.279	40.488	-2.950
	(0.088)	(0.094)	(0.124)	(23.411)	(27.612)	(26.519)
Video + Sav. Cal.	0.172*	0.151	0.084	39.201*	36.069	-17.995
	(0.096)	(0.104)	(0.127)	(22.855)	(24.809)	(27.888)
Saving for old age (Bsl.)	0.334***	0.333***	0.352***			
	(0.079)	(0.079)	(0.103)			
Female		0.044			15.773	
		(0.189)			(40.474)	
Female × (Video + Plac. Cal.)		0.133			-66.336	
		(0.241)			(48.625)	
Female × (Video + Sav. Cal.)		0.140			5.793	
		(0.278)			(63.243)	
Total savings (Bsl.)			-0.024*			1.423
			(0.014)			(3.993)
Total savings (Bsl.) × (Video + Plac. Cal.)			0.014			7.056
			(0.027)			(15.894)
Total savings (Bsl.) × (Video + Sav. Cal.)			0.057			42.971**
			(0.035)			(17.872)
Constant	1.642***	1.673***	1.765***	35.637	29.936	93.168
	(0.239)	(0.244)	(0.322)	(47.285)	(47.250)	(64.151)
Baseline controls	Yes	Yes	Yes	Yes	Yes	Yes
Observations	618	618	367	235	235	153
R-squared	0.051	0.051	0.065	0.027	0.034	0.170

Notes: Columns 1-3 of this table present the results of an ANCOVA specification to assess treatments effects, and the heterogeneous treatment effects by gender (Female) and by total savings at baseline (Total savings (Bsl.)) on savings for old age. Columns 4-6 of this table present the results of a post-treatment regression specification to assess treatment effects, and the heterogeneous treatment effects by gender (Female) and by total savings at baseline (Total savings (Bsl.)) on PPT pension savings (recorded in Ghana cedis). The treatment variables are video plus placebo calendar (Vid. + Plac. Cal.) and video plus savings calendar (Vid. + Sav. Cal.). Baseline controls consist of the variables that were unbalanced at baseline (Female; Age; Migration status; Farm equipment; Social capital; Financial difficulty; Natural/biological disaster), excluding Female in Columns 2 and 5, and the attrition predictor (Salience of old age). See Table 3 for variable definitions. Full regressions showing coefficients of the baseline controls are presented in Appendix Table C.8. Huber-White robust standard errors are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

employ a Heckman regression model. This approach allows us to assess whether the unobserved factors influencing the selection process (i.e., responding to the PPT pension savings question) are significantly related to the unobserved factors affecting PPT pension savings.

The results indicate no significant correlation between the error terms of the selection and outcome equations (see Appendix Table E.1). Furthermore, including significant predictors of responding to the PPT pension savings question from the Heckman selection model does not substantially alter the estimated treatment effects (see Appendix Table E.2), underlining the robustness of our findings.

We further explore heterogeneity in the treatment effect on PPT pension savings by gender and total savings at baseline. We find no heterogeneous treatment effects by gender (see Column 5 of Table 8). However, we find that the treatment effect of the *Vid. + Sav. Cal.* on the average amount of pension savings is significantly higher among smallholders with higher initial levels of total savings. According to our estimates, these farmers saved approximately GH¢ 43 more (see Column 6). Thus, the treatment appears to have reinforced existing savings behavior: while the treatment effect on the extensive margin of savings (i.e., the fraction of farmers who save for old age, see Table 8) does not vary by baseline savings level, the intensive margin is stronger for those with higher baseline savings. This result mirrors our finding that higher initial levels of total savings predict larger marginal effects on aspirations and is once again reminiscent of Gehrke et al. (2023) regarding educational outcomes. They observed substantial heterogeneity in treatment effects with respect to baseline student performance: specifically, their intervention increased schooling efforts among high-performing students, while treated low-performing students reduced their educational investments.

6 Discussion and conclusion

This study evaluates the effectiveness of two randomized interventions designed to encourage smallholder cocoa farmers in Ghana to save more for their pensions: a video intended to enhance aspirations regarding financial well-being in old age and a savings calendar aimed at increasing the salience of the need to save for old age and specific savings goals to that end. The video intervention was grounded in the literature on aspirations and hope, in line with Ray (2006)'s idea of the aspirations window (i.e., the set of people whom one perceives to be similar to oneself, and who are used as reference points when forming aspirations). The savings calendar intervention, in turn, was intended to keep the video-induced aspirations salient in the longer run, informed by existing evidence about the importance of such salience (e.g., Karlan et al., 2016) as well as the possible low persistence of induced aspirations (Bernard et al., 2023). As such, the interventions aimed to address psychological barriers to long-term savings. Our findings provide important insights into how such interventions can influence savings behavior, and thus inform the design and rollout of effective pro-poor old-age welfare policies, particularly in contexts where formal social safety nets are weak or absent.

Our key results are that the video significantly increased aspirations in the short run, and that the combination of the video and the savings calendar led to a statistically significant increase in the fraction of farmers saving for old age in the long run. Farmers in this treatment group were 17.2 percentage points more likely to save for old age compared to the control group. The combination of the video and the savings calendar also led to a statistically significant increase in the average amount of pension savings, which was particularly pronounced among farmers with higher levels of savings at baseline, who increased their pension contributions by approximately ₵43 (about 85% of a day's wage for a farm laborer at the time of the study). This is consistent with findings from related literature (e.g., Gehrke et al., 2023), where interventions tend to reinforce existing positive behaviors rather than uniformly benefiting all participants.

While the video successfully raised aspirations in the short term, its effects on aspirations faded over time, in line with patterns observed in other studies (e.g., Rojas Valdes et al., 2022), and were not statistically significant on average in the long run. However, these effects were not evenly distributed across different groups: the marginal effect of the video in combination with the savings calendar on aspirations was lower for women, while the marginal effects of the video, both in combination with the placebo calendar and with the savings calendar, were larger among farmers with higher initial total savings.

Similarly, the combination of the video and the savings calendar benefited women less than men in terms of its long-term effects on agency and hope, although its impact on participation and the amount of pension savings did not differ between women and men. This suggests that the rigid commitment implied by writing down savings goals in the savings calendar may have backfired in terms of agency and hope for women, possibly because they face greater structural constraints (e.g., limited access to resources and decision-making power).

The finding that farmers with higher initial savings responded more strongly to the intervention implies that one-size-fits-all approaches may not be optimal. Policymakers and practitioners could therefore consider complementary support measures for individuals with lower baseline savings, for example, financial literacy training or improved access to credit, to help address structural barriers.

The video intervention as part of either treatment had a positive long-term effect on hope and agency, suggesting that aspirational messaging that aligns with one's aspiration window (in our case, positive experiences about pension savings narrated by a person close to the study subjects' socio-economic and demographic description) can foster a sense of control over one's financial future. While the video affected agency and hope in the long run, increased savings were observed for the treatment that paired it with a tool that kept the induced aspirations salient over time (the savings calendar).

The savings calendar's success in supporting savings along both the extensive and intensive margins suggests that visual and tangible tools can help bridge the gap between intentions and actions. To sustain long-term behavioral change, future interventions could incorporate ongoing engagement mechanisms, such as periodic reminders, peer support groups, or adaptive goal-setting tools.

The negative impact on women's sense of agency and hope is a critical concern. Future interventions may therefore benefit from gender-sensitive adaptations, such as flexible goal-setting (e.g., allowing adjustments for unforeseen shocks), targeted messaging that addresses women's unique constraints (e.g., lack of control over household finances), and/or complementary resources (e.g., access to women's savings groups or microcredit).

The study suggests that psychological interventions alone may not be sufficient to drive lasting behavioral changes in all segments of the population. While the combination of the video and savings calendar proved effective in increasing pension savings participation, its heterogeneous effects, especially by gender, suggest that interventions are more likely to be uniformly successful when they address both internal (e.g., aspirations, agency) and external (e.g., gender disparities) barriers.

While this study provides valuable insights, the relatively small sample size may limit the broader applicability of the findings. Future research should explore similar interventions with larger samples. Besides, the study period coincided with the COVID-19 pandemic, which may have disproportionately affected more vulnerable groups within the sample, such as women. Longitudinal studies could examine how shocks such as the COVID-19 pandemic interact with intervention effects over time, particularly for more vulnerable groups.

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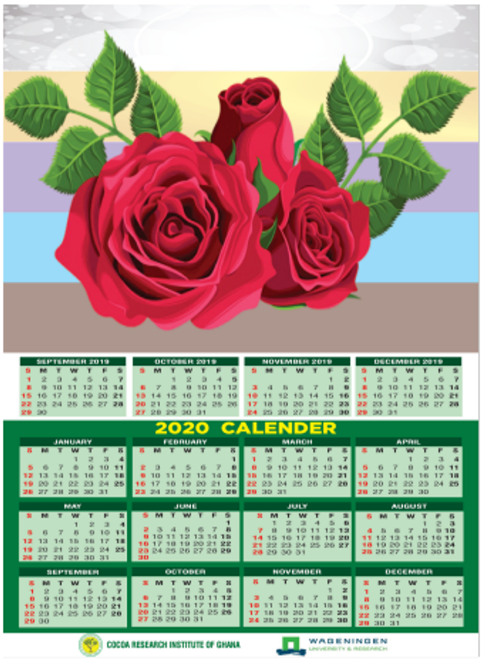
Appendices

A Appendix: experiment details

A.1 Calendar treatments

Figure A.1 depicts the placebo as well as the treatment (savings) calendars.



(a) Placebo calendar

(b) Savings calendar

Figure A.1: "Placebo calendar" (left frame) and "Savings calendar" (right frame).

A.2 Video treatment

Below we provide the content of our treatment video, while Figure A.2 provides a screenshot of the video.

[Scene 1.]

It is daytime in what appears to be a rural setting (seeds being dried, a farm animal, etc.). An older man ("pension recipient") speaks while walking about in front of a house that looks well-maintained. The pension recipient is dressed in clean traditional garments and has other demographic and socio-economic characteristics comparable to those of the smallholder farmers involved in the study. Later in the scene, he indicates towards another old man (the uncle referred to in the transcript), who sits in front of a smaller house and appears to be in a more penurious situation, indicated by torn,

Figure A.2: Screenshot from the video (Scene 2)

mismatched footwear. The uncle nods and shakes his head at times while being spoken about by the pension recipient.

[Narration in Scene 1.]

Pension recipient: I remember when I was a young man. I had a cocoa farm. It wasn't a big farm, so the funds I had from selling my produce weren't much. My uncle advised me to save for my pension, irrespective of how small my contribution was, so I could depend on it later in life. I adhered to his advice. The money wasn't much but I contributed the little I had diligently.
[Continuing to speak while indicating at his uncle:] I had another uncle who did not heed this advice. He doesn't have decent accommodation. Even his roof leaks badly. He is old now and can't work anymore, but since he didn't save for his retirement, he has fallen on hard times. I, on the other hand, have been able to build my house and cater to my kids. If I hadn't saved for my retirement, I don't know how I would fend for myself in my old age.

[Scene 2.]

It is daytime in a well-lit room where farmers, including men and women of different ages, are seated. A group meeting (cocoa cooperative) is in progress and the discussion is about old age and pensions. The group leader, an older man, speaks first. A young man asks a question. The pension recipient from the previous scene gets up and answers. When he finishes speaking, an older woman provides her views, followed by a second older woman.

[Narration in Scene 2.]

Group leader: Cocoa – Ghana, Ghana – Cocoa, Koo, you and I.
[Group claps]
Group leader: It is very important that as farmers, we save for our retirement, especially the youth.

Young man raises hand: Secretary, I want to ask a question. I want to know why you emphasized the need for the youth to save for their retirement. We are quite young now and we have a number of things to do with our money. So I don't understand why the youth are being entreated to save for an event that might happen later in life.

Pension recipient from earlier scene stands up and speaks: Young man, I was once like you. If I hadn't saved as a young man, I would have been miserable by now. How would I have provided for my children's education, food, and other basic needs? The little funds I saved from selling my farm produce are what I used for my house in Accra and even the one here in Heman. I beg you to reconsider your decision and save some money for your pension. You will definitely grow old, and even fall sick, and not have enough energy to work. Saving for your retirement is essential.

Old lady 1: It is important you save for your pension. It has helped me in my old age. The money I saved during my youth is what I have relied on till date. When my uncle was advising me to save for my retirement, he told me about another man who hadn't saved at all. Today, he doesn't have a proper roof over his head. No matter how difficult things get, try to save for your retirement. It is really important.

Old lady 2: I agree with everything that has been said here. I am a farmer. My husband died about 7 years ago. He was also a farmer with seasonal income. He thought about the nature of our earnings and realized that a pension would be beneficial for us in the future. So, he joined a pension scheme and listed me as a beneficiary. He also advised me to join one. Upon his demise, his pension was paid to me, and I also receive my own now that I am retired. We buy farming supplies from that too. I'd advise everyone to get enrolled in a pension scheme. A pension scheme is not only for men. Women should join a pension scheme too and save for their retirement.

B Appendix: attrition and balance tests

Table B.1: Attrition frequency

Attrition	Freq.	Percent
Present at baseline	723	100
Of whom present at endline	620	85.75
Of whom present only at baseline	103	14.25

Notes: The table presents the breakup of the number of farmers who participated in the baseline into those who were also present at the endline survey and those who were not (i.e., were present only at the baseline).

Table B.2: Attrition test by treatment status

Variables	Attrition
Vid. + Plac. Cal.	-0.028
	(0.032)
Vid. + Sav. Cal.	-0.029
	(0.032)
Constant	0.162***
	(0.023)
Observations	723
R-squared	0.001
F(2, 720)	0.53

Notes: The table presents the results of an attrition test using the treatment variables video plus placebo calendar (*Vid. + Plac. Cal.*) and video plus savings calendar (*Vid. + Sav. Cal.*) as explanatory variables. Standard errors are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table B.3: Attrition test on treatments, baseline outcomes and demographics

Variables	Attrition	Variables	Attrition
Vid. + Plac. Cal.	-0.022	Income	0.000
	(0.037)		(0.000)
Vid. + Sav. Cal.	-0.041	Total savings (Bsl.)	-0.002
	(0.036)		(0.005)
Aspirations (Bsl.)	-0.015	Saving for old age	0.027
	(0.018)		(0.032)
Agency (Bsl.)	0.010	Total farm size	-0.001
	(0.018)		(0.001)
Pathways (Bsl.)	0.001	Farm equipment	0.024
	(0.018)		(0.023)
Salience of old age	-0.428***	House size	0.001
	(0.032)		(0.006)
Female	0.004	Elec. and electr. assets	-0.005
	(0.042)		(0.011)
Age	-0.001	Vehicles	0.033
	(0.001)		(0.025)
Household size	-0.001	Leadership status	-0.003
	(0.006)		(0.043)
Migration status	-0.006	Social capital	0.003
	(0.034)		(0.013)
Speaks Twi	-0.040	Financial difficulty	0.039
	(0.041)		(0.037)
Education	-0.004	Natural/biological disaster	0.013
	(0.006)		(0.031)
Financial literacy	0.014	Constant	0.539***
	(0.033)		(0.112)
		Observations	387
		R-squared	0.357

Notes: The table presents the results of an attrition test using the treatment variables video plus placebo calendar (Vid. + Plac. Cal.) and video plus savings calendar (Vid. + Sav. Cal.), the psychological outcome variables (Aspirations, Agency, and Pathways) at baseline, and control variables as explanatory variables. See Table 3 for variable definitions. Standard errors are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table B.4: Balance table

Categories	Variables	Obs.	(1) Control	(2) Vid. + Plac. Cal.	(3) Vid. + Sav. Cal.	(1) vs. (2)	(1) vs. (3)	(2) vs. (3)
Psychological outcome variables	Aspirations (Bsl.)	723	0.000	0.049	0.068	-0.049	-0.068	-0.020
			(0.065)	(0.052)	(0.066)	(0.083)	(0.093)	(0.084)
	Agency (Bsl.)	723	0.000	0.145	0.121	-0.145	-0.121	0.024
			(0.065)	(0.065)	(0.066)	(0.092)	(0.093)	(0.093)
	Pathways (Bsl.)	723	0.000	-0.050	0.131	0.050	-0.131	-0.181*
			(0.065)	(0.078)	(0.064)	(0.102)	(0.091)	(0.100)
Old salience	Salience of old age	723	0.660	0.675	0.698	-0.015	-0.038	-0.023
			(0.031)	(0.030)	(0.029)	(0.043)	(0.043)	(0.042)
Demographics	Female	723	0.145	0.250	0.194	-0.105***	-0.049	0.056
			(0.023)	(0.028)	(0.025)	(0.036)	(0.034)	(0.038)
	Age	721	57.043	55.651	54.923	1.391	2.119*	0.728
			(0.811)	(0.838)	(0.777)	(1.167)	(1.123)	(1.142)
	Household size	723	5.723	5.721	5.540	0.003	0.183	0.181
			(0.179)	(0.153)	(0.144)	(0.235)	(0.228)	(0.209)
	Migration status	722	0.329	0.342	0.258	-0.013	0.071*	0.084**
			(0.031)	(0.031)	(0.028)	(0.043)	(0.041)	(0.041)
	Speaks Twi	723	0.826	0.808	0.823	0.017	0.003	-0.014
			(0.025)	(0.025)	(0.024)	(0.036)	(0.035)	(0.035)
	Education	705	9.180	9.110	9.527	0.070	-0.347	-0.417
			(0.180)	(0.199)	(0.174)	(0.269)	(0.250)	(0.264)
	Financial literacy	723	0.617	0.633	0.673	-0.016	-0.056	-0.040
			(0.032)	(0.031)	(0.030)	(0.045)	(0.044)	(0.043)
Income and savings	Income	662	8.073	7.907	17.409	0.166	-9.336	-9.502
			(0.549)	(0.566)	(7.936)	(0.790)	(8.323)	(8.097)
	Total savings (Bsl.)	423	1.311	1.260	1.353	0.051	-0.042	-0.093
			(0.295)	(0.198)	(0.214)	(0.359)	(0.359)	(0.295)
	Saving for old age	723	0.600	0.575	0.536	0.025	0.064	0.039
			(0.032)	(0.032)	(0.032)	(0.045)	(0.045)	(0.045)
Other assets	Total farm size	717	8.472	10.871	8.679	-2.399	-0.207	2.192
			(1.031)	(1.314)	(1.371)	(1.672)	(1.729)	(1.902)
	Farm equipment	723	0.864	0.887	0.786	-0.024	0.078	0.101*
			(0.043)	(0.043)	(0.042)	(0.061)	(0.060)	(0.060)
	House size	723	4.064	4.025	4.246	0.039	-0.182	-0.221
			(0.159)	(0.144)	(0.161)	(0.215)	(0.227)	(0.216)
	Elec. and electr. assets	723	3.196	3.233	3.157	-0.038	0.038	0.076
			(0.106)	(0.105)	(0.094)	(0.149)	(0.141)	(0.140)
	Vehicles	723	0.502	0.429	0.423	0.073	0.079	0.006
			(0.044)	(0.042)	(0.040)	(0.060)	(0.059)	(0.057)
Social standing	Leadership status	723	0.149	0.150	0.177	-0.001	-0.028	-0.027
			(0.023)	(0.023)	(0.024)	(0.033)	(0.034)	(0.034)
	Social capital	723	2.123	2.112	2.306	0.011	-0.183*	-0.194*
			(0.073)	(0.079)	(0.076)	(0.108)	(0.106)	(0.109)
Shocks	Financial difficulty	723	0.140	0.175	0.198	-0.035	-0.057*	-0.023
			(0.023)	(0.025)	(0.025)	(0.033)	(0.034)	(0.035)
	Natural/biological disaster	723	0.391	0.304	0.359	0.087**	0.033	-0.055
			(0.032)	(0.030)	(0.031)	(0.044)	(0.044)	(0.043)

Notes: The table presents balance tests. Variable definitions follow Table 3. Standard errors are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

C Appendix: additional regression results.

Table C.1: Long-run treatment effects on psychological outcome variables (without controls)

Variables	(1) Aspirations	(2) Hope	(3) Agency	(4) Pathways
Video + Plac. Cal.	0.014	0.150**	0.155**	0.070
	(0.081)	(0.073)	(0.075)	(0.068)
Video + Sav. Cal.	0.086	0.073	0.063	0.065
	(0.084)	(0.069)	(0.073)	(0.070)
Aspirations (Bsl.)	0.022			
	(0.039)			
Hope (Bsl.)		0.151***		
		(0.032)		
Agency (Bsl.)			0.154***	
			(0.030)	
Pathways (Bsl.)				0.052**
				(0.026)
Constant	-0.499***	-0.194***	-0.037	-0.346***
	(0.057)	(0.052)	(0.057)	(0.048)
Baseline controls	No	No	No	No
Observations	618	618	618	618
R-squared	0.003	0.044	0.054	0.009
Test of differences in coefficients of the two treatments				
F(1, 614)	0.74	1.27	1.90	0.01
Prob > F	0.391	0.261	0.169	0.942

Notes: The table presents the results of an ANCOVA specification to assess treatment effects on aspirations, hope, agency, and pathways. The treatment variables are video plus placebo calendar (Vid. + Plac. Cal.) and video plus savings calendar (Vid. + Sav. Cal.). Huber-White robust standard errors are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table C.2: Long-run heterogeneous treatment effects by gender on psychological outcome variables (without controls)

Variables	(1) Aspirations	(2) Hope	(3) Agency	(4) Pathways
Female	0.190	0.275*	0.313**	0.132
	(0.199)	(0.146)	(0.149)	(0.146)
Video + Plac. Cal.	0.014	0.151*	0.168**	0.060
	(0.089)	(0.081)	(0.085)	(0.076)
Video + Sav. Cal.	0.165*	0.147*	0.127	0.132*
	(0.089)	(0.075)	(0.081)	(0.076)
Female × (Video + Plac. Cal.)	-0.090	-0.135	-0.199	-0.025
	(0.237)	(0.188)	(0.185)	(0.186)
Female × (Video + Sav. Cal.)	-0.437*	-0.449**	-0.422**	-0.357*
	(0.252)	(0.188)	(0.186)	(0.196)
Aspirations (Bsl.)	0.017			
	(0.038)			
Hope (Bsl.)		0.144***		
		(0.032)		
Agency (Bsl.)			0.149***	
			(0.030)	
Pathways (Bsl.)				0.048*
				(0.025)
Constant	-0.523***	-0.228***	-0.077	-0.363***
	(0.059)	(0.056)	(0.062)	(0.051)
Baseline controls	No	No	No	No
Observations	618	618	618	618
R-squared	0.011	0.054	0.063	0.017
Lincom:				
(Video + Sav. Cal.) +	-0.302*	-0.311*	-0.295*	-0.225
Female × (Video + Sav. Cal.)	(0.172)	(0.170)	(0.168)	(0.181)

Notes: The table presents the results of an ANCOVA specification to assess the heterogeneous treatment effects by gender (Female) on aspirations, hope, agency, and pathways. The treatment variables are video plus placebo calendar (Vid. + Plac. Cal.) and video plus savings calendar (Vid. + Sav. Cal.). See Table 3 for variable definitions. Huber-White robust standard errors are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table C.3: Long-run heterogeneous treatment effects by initial level of total savings on psychological outcome variables (without controls)

Variables	(1) Aspirations	(2) Hope	(3) Agency	(4) Pathways
Total savings (Bsl.)	-0.025***	0.007	0.006	-0.001
	(0.007)	(0.010)	(0.011)	(0.006)
Video + Plac. Cal.	-0.063	0.168*	0.170*	0.060
	(0.116)	(0.098)	(0.103)	(0.086)
Video + Sav. Cal.	-0.078	0.014	-0.013	0.036
	(0.117)	(0.100)	(0.106)	(0.092)
Total savings (Bsl.) × (Video + Plac. Cal.)	0.070**	0.011	0.036*	-0.013
	(0.030)	(0.020)	(0.021)	(0.017)
Total savings (Bsl.) × (Video + Sav. Cal.)	0.072**	0.035	0.036	0.020
	(0.035)	(0.038)	(0.042)	(0.029)
Aspirations (Bsl.)	-0.003			
	(0.042)			
Hope (Bsl.)		0.140***		
		(0.039)		
Agency (Bsl.)			0.171***	
			(0.038)	
Pathways (Bsl.)				0.017
				(0.028)
Constant	-0.416***	-0.190***	-0.041	-0.313***
	(0.078)	(0.068)	(0.077)	(0.055)
Baseline controls	No	No	No	No
Observations	367	367	367	367
R-squared	0.013	0.054	0.091	0.005
Lincom:				
(Video + Sav. Cal.) + Total savings (Bsl.)	-0.006	0.049	0.023	0.056
× (Video + Sav. Cal.)	(0.107)	(0.088)	(0.093)	(0.082)

Notes: The table presents the results of an ANCOVA specification to assess the heterogeneous treatment effects by total savings (Total savings (Bsl.)) on aspirations, hope, agency, and pathways. The treatment variables are video plus placebo calendar (Vid. + Plac. Cal.) and video plus savings calendar (Vid. + Sav. Cal.). See Table 3 for variable definitions. Huber-White robust standard errors are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table C.4: Long-run treatment effects on savings outcomes (without controls)

	Saving for old age (Endl.)			PPT pension savings		
Variables	(1)	(2)	(3)	(4)	(5)	(6)
Video + Plac. Cal.	0.018	-0.014	0.131	26.039	38.337	12.386
	(0.087)	(0.094)	(0.121)	(22.701)	(27.906)	(25.289)
Video + Sav. Cal.	0.195**	0.162	0.128	37.405*	34.683	-13.040
	(0.096)	(0.102)	(0.129)	(21.471)	(23.544)	(27.861)
Saving for old age (Bsl.)	0.340***	0.345***	0.355***			
	(0.077)	(0.077)	(0.098)			
Female		0.003			11.083	
		(0.187)			(34.964)	
Female × (Video + Plac. Cal.)		0.128			-61.308	
		(0.243)			(44.825)	
Female × (Video + Sav. Cal.)		0.154			9.283	
		(0.274)			(60.554)	
Total savings (Bsl.)			-0.020			3.616
			(0.013)			(2.413)
Total savings (Bsl.) × (Video + Plac. Cal.)			0.015			1.249
			(0.025)			(14.422)
Total savings (Bsl.) × (Video + Sav. Cal.)			0.045			38.905**
			(0.034)			(17.636)
Constant	1.807***	1.804***	1.827***	71.768***	70.417***	66.512***
	(0.080)	(0.084)	(0.099)	(10.069)	(10.564)	(14.547)
Baseline controls	No	No	No	No	No	No
Observations	618	618	367	235	235	153
R-squared	0.038	0.041	0.045	0.011	0.019	0.139

Notes: Columns 1-3 of this table present the results of an ANCOVA specification to assess treatment effects, and the heterogeneous treatment effects by gender (Female) and by total savings at baseline (Total savings (Bsl.)) on savings for old age. Columns 4-6 of this table present the results of a post-treatment regression specification to assess treatment effects, and the heterogeneous treatment effects by gender (Female) and by total savings at baseline (Total savings (Bsl.)) on PPT pension savings (recorded in Ghana cedis). The treatment variables are video plus placebo calendar (Vid. + Plac. Cal.) and video plus savings calendar (Vid. + Sav. Cal.). See Table 3 for variable definitions. Huber-White robust standard errors are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table C.5: Short- and long-run treatment effects on psychological outcome variables (including control coefficients)

Variables	(1) Aspirations (Mdl.)	(2) Aspirations (Endl.)	(3) Hope (Endl.)	(4) Agency (Endl.)	(5) Pathways (Endl.)
Video	0.120*				
	(0.070)				
Video + Plac. Cal.		0.017	0.142**	0.147**	0.064
		(0.082)	(0.072)	(0.075)	(0.068)
Video + Sav. Cal.		0.071	0.058	0.042	0.057
		(0.084)	(0.068)	(0.073)	(0.070)
Aspirations (Bsl.)	0.518***	0.012			
	(0.075)	(0.039)			
Hope (Bsl.)			0.132***		
			(0.030)		
Agency (Bsl.)				0.147***	
				(0.030)	
Pathways (Bsl.)					0.033
					(0.024)
Salience of old age	0.013	0.273***	0.229***	0.207***	0.182***
	(0.087)	(0.080)	(0.067)	(0.075)	(0.064)
Female	0.069	0.007	0.101	0.114	0.040
	(0.091)	(0.089)	(0.073)	(0.070)	(0.078)
Age	-0.003	-0.001	-0.004*	-0.003	-0.005**
	(0.003)	(0.003)	(0.002)	(0.003)	(0.002)
Migration status	-0.137*	0.020	-0.005	-0.078	0.101
	(0.071)	(0.072)	(0.064)	(0.068)	(0.062)
Farm equipment	-0.054	0.039	0.120***	0.111**	0.090**
	(0.066)	(0.053)	(0.044)	(0.044)	(0.045)
Social capital	-0.016	0.076**	0.033	0.030	0.036
	(0.027)	(0.030)	(0.024)	(0.025)	(0.024)
Financial difficulty	-0.012	-0.003	0.048	0.004	0.109
	(0.084)	(0.088)	(0.077)	(0.076)	(0.081)
Natural shocks	-0.213***	-0.069	-0.028	-0.063	0.018
	(0.082)	(0.070)	(0.059)	(0.062)	(0.059)
Constant	0.382*	-0.844***	-0.335**	-0.181	-0.407***
	(0.195)	(0.196)	(0.164)	(0.177)	(0.149)
Observations	618	618	618	618	618
R-squared	0.282	0.035	0.085	0.088	0.048

Notes: The table presents the results of an ANCOVA specification to assess the treatment effect on short-run aspirations in Column 1 and treatment effects on long-run aspirations, hope, agency, and pathways in Columns 2, 3, 4, and 5. The treatment variables are video (Video) in Column 1 and video plus placebo calendar (Vid. + Plac. Cal.) as well as video plus savings calendar (Vid. + Sav. Cal.) in Columns 2, 3, 4, and 5. Controls consist of the variables that were unbalanced at baseline (Female; Age; Migration status; Farm equipment; Social capital; Financial difficulty; Natural/biological disaster) and the attrition predictor (Salience of old age). See Table 3 for variable definitions. Huber-White robust standard errors are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table C.6: Long-run heterogeneous treatment effects by gender on psychological outcome variables (including control coefficients)

Variables	(1) Aspirations (Endl.)	(2) Hope (Endl.)	(3) Agency (Endl.)	(4) Pathways (Endl.)
Female	0.224	0.318**	0.346**	0.175
	(0.192)	(0.148)	(0.149)	(0.149)
Video + Plac. Cal.	0.019	0.146*	0.166**	0.050
	(0.089)	(0.080)	(0.084)	(0.075)
Video + Sav. Cal.	0.154*	0.139*	0.116	0.122
	(0.090)	(0.075)	(0.081)	(0.076)
Female × (Video + Plac. Cal.)	-0.111	-0.119	-0.181	-0.010
	(0.229)	(0.188)	(0.181)	(0.190)
Female × (Video + Sav. Cal.)	-0.461*	-0.451**	-0.422**	-0.347*
	(0.242)	(0.187)	(0.184)	(0.198)
Aspirations (Bsl.)	0.007			
	(0.039)			
Hope (Bsl.)		0.125***		
		(0.030)		
Agency (Bsl.)			0.142***	
			(0.030)	
Pathways (Bsl.)				0.029
				(0.024)
Salience of old age	0.275***	0.231***	0.209***	0.184***
	(0.080)	(0.067)	(0.075)	(0.064)
Age	-0.000	-0.003	-0.002	-0.005**
	(0.003)	(0.002)	(0.002)	(0.002)
Migration status	0.020	-0.005	-0.078	0.101
	(0.073)	(0.064)	(0.068)	(0.062)
Farm equipment	0.047	0.126***	0.116***	0.096**
	(0.053)	(0.044)	(0.044)	(0.045)
Social capital	0.077**	0.034	0.031	0.036
	(0.030)	(0.024)	(0.025)	(0.024)
Financial difficulty	-0.001	0.051	0.006	0.111
	(0.089)	(0.076)	(0.076)	(0.081)
Natural shocks	-0.069	-0.029	-0.063	0.016
	(0.070)	(0.059)	(0.062)	(0.059)
Constant	-0.920***	-0.409**	-0.256	-0.458***
	(0.194)	(0.168)	(0.180)	(0.155)
Observations	618	618	618	618
R-squared	0.043	0.095	0.096	0.057

Notes: The table presents the results of an ANCOVA specification to assess the heterogeneous treatment effects by gender (Female) on aspirations, hope, agency, and pathways. The treatment variables are video plus placebo calendar (Vid. + Plac. Cal.) and video plus savings calendar (Vid. + Sav. Cal.). Controls consist of the variables that were unbalanced at baseline (Age; Migration status; Farm equipment; Social capital; Financial difficulty; Natural/biological disaster), excluding Female, and the attrition predictor (Salience of old age). See Table 3 for variable definitions. Huber-White robust standard errors are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table C.7: Long-run heterogeneous treatment effects by initial level of total savings on psychological outcome variables (including control coefficients)

Variables	(1) Aspirations (Endl.)	(2) Hope (Endl.)	(3) Agency (Endl.)	(4) Pathways (Endl.)
Total savings (Bsl.)	-0.023***	0.003	0.004	-0.006
	(0.008)	(0.009)	(0.010)	(0.006)
Video + Plac. Cal.	-0.048	0.168*	0.166	0.071
	(0.119)	(0.096)	(0.101)	(0.086)
Total savings (Bsl.) × (Video + Plac. Cal.)	0.072**	0.013	0.030	-0.006
	(0.032)	(0.021)	(0.022)	(0.018)
Video + Sav. Cal.	-0.085	-0.007	-0.047	0.041
	(0.121)	(0.100)	(0.104)	(0.093)
Total savings (Bsl.) × (Video + Sav. Cal.)	0.086**	0.049	0.047	0.033
	(0.035)	(0.036)	(0.040)	(0.029)
Aspirations (Bsl.)	-0.002			
	(0.043)			
Hope (Bsl.)		0.136***		
		(0.038)		
Agency (Bsl.)			0.164***	
			(0.038)	
Pathways (Bsl.)				0.017
				(0.028)
Female	-0.079	0.149	0.182**	0.025
	(0.128)	(0.093)	(0.086)	(0.097)
Age	0.007*	-0.002	-0.004	0.001
	(0.004)	(0.003)	(0.003)	(0.003)
Migration status	-0.025	0.040	-0.008	0.079
	(0.099)	(0.085)	(0.088)	(0.080)
Salience of old age	0.231**	0.233***	0.212**	0.152**
	(0.114)	(0.088)	(0.099)	(0.076)
Farm equipment	0.009	0.085	0.074	0.078
	(0.071)	(0.055)	(0.054)	(0.055)
Social capital	0.030	0.005	0.023	-0.013
	(0.040)	(0.034)	(0.035)	(0.030)
Financial difficulty	0.082	0.126	0.079	0.170*
	(0.113)	(0.092)	(0.091)	(0.092)
Natural/biological disaster	-0.041	-0.021	-0.094	0.091
	(0.092)	(0.073)	(0.076)	(0.071)
Constant	-1.062***	-0.399*	-0.098	-0.606***
	(0.281)	(0.221)	(0.238)	(0.192)
Observations	367	367	367	367
R-squared	0.036	0.092	0.127	0.040

Notes: The table presents the results of an ANCOVA specification to assess the heterogeneous treatment effects by total savings at baseline (Total savings (Bsl.)) on aspirations, hope, agency, and pathways. The treatment variables are video plus placebo calendar (Vid. + Plac. Cal.) and video plus savings calendar (Vid. + Sav. Cal.). Controls consist of the variables that were unbalanced at baseline (Female; Age; Migration status; Farm equipment; Social capital; Financial difficulty; Natural/biological disaster) and the attrition predictor (Salience of old age). See Table 3 for variable definitions. Huber-White robust standard errors are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table C.8: Long-run treatment effects on savings outcomes (including control coefficients)

	Saving for old age (Endl.)			PPT pension savings		
Variables	(1)	(2)	(3)	(4)	(5)	(6)
Video + Plac. Cal.	0.004	-0.017	0.110	28.279	40.488	-2.950
	(0.088)	(0.094)	(0.124)	(23.411)	(27.612)	(26.519)
Video + Sav. Cal.	0.172*	0.151	0.084	39.201*	36.069	-17.995
	(0.096)	(0.104)	(0.127)	(22.855)	(24.809)	(27.888)
Female	0.150	0.044	0.300**	-7.963	15.773	12.059
	(0.107)	(0.189)	(0.148)	(23.584)	(40.474)	(32.175)
Female × (Video + Plac. Cal.)		0.133			-66.336	
		(0.241)			(48.625)	
Female × (Video + Sav. Cal.)		0.140			5.793	
		(0.278)			(63.243)	
Total savings (Bsl.)			-0.024*			1.423
			(0.014)			(3.993)
Total savings (Bsl.) × (Video + Plac. Cal.)			0.014			7.056
			(0.027)			(15.894)
Total savings (Bsl.) × (Video + Sav. Cal.)			0.057			42.971**
			(0.035)			(17.872)
Saving for old age (Bsl.)	0.334***	0.333***	0.352***			
	(0.079)	(0.079)	(0.103)			
Salience of old age (Bsl.)	0.051	0.051	-0.047	10.078	9.958	33.092
	(0.103)	(0.103)	(0.118)	(21.461)	(21.439)	(43.019)
Age	0.0004	0.0001	-0.002	0.079	0.113	-0.015
	(0.003)	(0.003)	(0.004)	(0.653)	(0.645)	(1.020)
Migration status	-0.107	-0.107	0.300**	34.861	34.356	12.059
	(0.083)	(0.083)	(0.148)	(26.881)	(26.713)	(32.175)
Farm equipment	0.101*	0.100*	0.044	-3.277	-4.041	11.034
	(0.058)	(0.059)	(0.134)	(16.275)	(16.499)	(24.226)
Social capital	0.017	0.016	0.097	4.689	5.382	-14.221
	(0.034)	(0.035)	(0.076)	(7.128)	(7.241)	(16.002)
Financial difficulty	0.118	0.118	0.016	6.198	5.512	-14.270
	(0.113)	(0.113)	(0.045)	(30.792)	(30.801)	(9.129)
Natural shocks	-0.046	-0.048	0.089	16.010	17.285	13.471
	(0.079)	(0.079)	(0.145)	(20.493)	(20.576)	(41.763)
Constant	1.642***	1.673***	1.765***	35.637	29.936	93.168
	(0.239)	(0.244)	(0.322)	(47.285)	(47.250)	(64.151)
Observations	618	618	367	235	235	153
R-squared	0.051	0.051	0.065	0.027	0.034	0.170

Notes: Columns 1-3 of this table present the results of an ANCOVA specification to assess treatment effects, and the heterogeneous treatment effects by gender (Female) and by total savings at baseline (Total savings (Bsl.)) on savings for old age. Columns 4-6 of this table present the results of a post-treatment regression specification to assess treatment effects, and the heterogeneous treatment effects by gender (Female) and by total savings at baseline (Total savings (Bsl.)) on PPT pension savings (recorded in Ghana cedis). The treatment variables are video plus placebo calendar (Vid. + Plac. Cal.) and video plus savings calendar (Vid. + Sav. Cal.). Controls consist of the variables that were unbalanced at baseline (Female; Age; Migration status; Farm equipment; Social capital; Financial difficulty; Natural/biological disaster), excluding Female in Columns 2 and 5, and the attrition predictor (Salience of old age). See Table 3 for variable definitions. Huber-White robust standard errors are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

D Appendix: Marginal effect of savings calendar

Table D.1: Marginal effect of savings calendar on hope, agency and pathways

	Hope (Endl.)		Agency (Endl.)		Pathways (Endl.)	
Variables	(1)	(2)	(3)	(4)	(5)	(6)
AnyVid.	0.150**	0.142**	0.155**	0.147**	0.070	0.064
	(0.073)	(0.072)	(0.075)	(0.075)	(0.068)	(0.068)
Video + Sav. Cal.	-0.077	-0.084	-0.093	-0.105	-0.005	-0.007
	(0.068)	(0.067)	(0.067)	(0.067)	(0.071)	(0.070)
Hope (Bsl.)	0.151***	0.132***				
	(0.032)	(0.030)				
Agency (Bsl.)			0.154***	0.147***		
			(0.030)	(0.030)		
Pathways (Bsl.)					0.052**	0.033
					(0.026)	(0.024)
Salience of old age		0.229***		0.207***		0.182***
		(0.067)		(0.075)		(0.064)
Female		0.101		0.114		0.040
		(0.073)		(0.070)		(0.078)
Age		-0.004*		-0.003		-0.005**
		(0.002)		(0.003)		(0.002)
Migration status		-0.005		-0.078		0.101
		(0.064)		(0.068)		(0.062)
Farm equipment		0.120***		0.111**		0.090**
		(0.044)		(0.044)		(0.045)
Social capital		0.033		0.030		0.036
		(0.024)		(0.025)		(0.024)
Financial difficulty		0.048		0.004		0.109
		(0.077)		(0.076)		(0.081)
Natural/biological disaster		-0.028		-0.063		0.018
		(0.059)		(0.062)		(0.059)
Constant	-0.194***	-0.335**	-0.037	-0.181	-0.346***	-0.407***
	(0.052)	(0.164)	(0.057)	(0.177)	(0.048)	(0.149)
Observations	618	618	618	618	618	618
R-squared	0.044	0.085	0.054	0.088	0.009	0.048

Notes: The table presents the results of an ANCOVA specification to assess the marginal effects of having watched the video as part of either treatment group and of the savings calendar over and above that effect on hope, agency and pathways. The treatment variables are any video treatment (AnyVid) and video plus savings calendar (Vid. + Sav. Cal.). Columns 1, 3, and 5 present estimates for hope, agency, and pathways, respectively, without controls. Columns 2, 4, and 6 present estimates for hope, agency, and pathways, respectively, with controls. Controls consist of the variables that were unbalanced at baseline (Female; Age; Migration status; Farm equipment; Social capital; Financial difficulty; Natural/biological disaster) and the attrition predictor (Salience of old age). See Table 3 for variable definitions. Huber-White robust standard errors are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table D.2: Marginal effect of savings calendar on PPT pension savings

Variables	(1) PPT pension savings
AnyVid.	28.279
	(23.411)
Video + Sav. Cal.	10.922
	(29.601)
Salience of old age	10.078
	(21.461)
Female	-7.963
	(23.584)
Age	0.079
	(0.653)
Migration status	34.861
	(26.881)
Farm equipment	-3.277
	(16.275)
Social capital	4.689
	(7.128)
Financial difficulty	6.198
	(30.792)
Natural/biological disaster	16.010
	(20.493)
Constant	35.637
	(47.285)
Observations	235
R-squared	0.027

Notes: The table presents the results of a post-treatment regression specification to assess the marginal effects of having watched the video as part of either treatment group and of the savings calendar over and above that effect on PPT pension savings (recorded in Ghana cedis). The treatment variables are any video treatment (AnyVid) and video plus savings calendar (Vid. + Sav. Cal.). Controls consist of the variables that were unbalanced at baseline (Female; Age; Migration status; Farm equipment; Social capital; Financial difficulty; Natural/biological disaster) and the attrition predictor (Salience of old age). See Table 3 for variable definitions. Huber-White robust standard errors are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

E Appendix: Robustness check for PPT pension savings results using Heckman

In our post-treatment regression to assess long-run treatment effects on PPT pension savings, we find that Vid. + Sav. Cal. is marginally significant (see Table 8). However, due to non-responses, the number of observations for the PPT pension savings outcome is substantially reduced. To examine whether non-response is systematically related to the outcome variable, we employ a Heckman selection model. This approach allows us to test whether unobserved factors influencing the selection process (i.e., responding to the PPT pension savings question) are correlated with unobserved factors affecting PPT pension savings. In the selection equation, we include all variables from the descriptive statistics (see Table 3), except for income, total farm size, and education, which prevent convergence.

The Heckman regression results show no statistically significant correlation between the error terms of the selection and outcome equations (see Column 3 of Appendix Table E.1), suggesting no evidence of selection bias; however, the treatment effects are not statistically significant in this specification.

Building on the selection results, we augment the baseline specification from Table 8 by including significant predictors from the first-stage selection equation that are not already part of our baseline controls (namely saving for old age and house size). The estimated treatment effect remains similar in magnitude and retains its marginal statistical significance after the inclusion of these predictors (see Column 1 of Appendix Table E.2). We also find similar results for the heterogeneous treatment effects when adding these predictors to the control set (see Columns 2 and 3 of Appendix Table E.2).

Table E.1: Treatment effects on PPT pension savings — Heckman regression model

Variables	(1) PPT pension savings	(2) PPT savings dummy	(3)
Vid. + Plac. Cal.	13.751		
	(17.288)		
Vid. + Sav. Cal.	16.563		
	(16.223)		
Salience of old age (Bsl.)	7.624	-0.189	
	(16.123)	(0.135)	
Female	6.714	-0.090	
	(18.551)	(0.149)	
Age	-0.245	-0.010**	
	(0.519)	(0.005)	
Household size		0.015	
		(0.023)	
Migration status	18.198	0.041	
	(19.255)	(0.123)	
Speaks Twi		-0.080	
		(0.149)	
Education		-0.014	
		(0.022)	
Financial literacy		0.157	
		(0.123)	
Income		0.001	
		(0.000)	
Saving for old age (Bsl.)		0.342***	
		(0.123)	
Total farm size		0.005	
		(0.003)	
Farm equipment	-0.502	-0.083	
	(12.336)	(0.092)	
House size		-0.078***	
		(0.025)	
Elec. and electr. assets		-0.039	
		(0.041)	
Vehicles		-0.037	
		(0.097)	
Leadership status		-0.126	
		(0.158)	
Social capital	4.822	0.103**	
	(5.122)	(0.049)	
Financial difficulty	4.403	-0.087	
	(22.034)	(0.150)	
Natural/biological disaster	11.825	-0.071	
	(16.103)	(0.120)	
Athrho			-0.041
			(0.181)
Lnsigma			4.899***
			(0.118)
Constant	44.075	0.984**	
	(36.054)	(0.420)	
Observations	556	556	556

Notes: The table presents the results of a Heckman regression to assess treatments effects on PPT pension savings. The treatment variables are video plus placebo calendar (Vid. + Plac. Cal.) and video plus savings calendar (Vid. + Sav. Cal.). Column 1 presents the second stage (i.e., the outcome model), with controls that consist of the variables that were unbalanced at baseline (Female; Age; Migration status; Farm equipment; Social capital; Financial difficulty; Natural/biological disaster) and the attrition predictor (Salience of old age). See Table 3 for variable definitions. Column 2 presents the first stage (i.e., the selection model), which includes all control variables of Table 3. See Table 3 for variable definitions. Column 3 presents the correlation coefficient between the selection and the outcome model and the variability in the PPT pension savings. Athrho suggests minimal correlation between the selection process and the outcome, Lnsigma indicates significant and substantial variability in the PPT pension savings of the respondents. Huber-White robust standard errors are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table E.2: Post-treatment regression estimates including predictors from the Heckman selection model

Variables	(1)	(2)	(3)
Vid. + Plac. Cal.	28.193	39.920	-0.097
	(23.874)	(28.310)	(26.269)
Vid. + Sav. Cal.	38.974*	34.954	-12.591
	(22.743)	(24.614)	(28.564)
Female	-5.419	14.690	13.095
	(24.540)	(43.394)	(31.598)
Female × Vid. + Plac. Cal.		-62.495	
		(51.887)	
Female × Vid. + Sav. Cal.		10.652	
		(64.580)	
Total savings (Bsl.)			-0.442
			(4.584)
Total savings (Bsl.) × Vid. + Plac. Cal.			5.464
			(15.105)
Total savings (Bsl.) × Vid. + Sav. Cal.			42.247**
			(18.332)
Salience of old age	8.823	9.027	14.122
	(23.250)	(23.277)	(25.249)
Age	-0.014	-0.005	-0.420
	(0.665)	(0.658)	(0.969)
Migration status	34.98	34.721	38.317
	(27.496)	(27.366)	(42.102)
Saving for old age	21.199	19.582	31.743
	(22.185)	(22.346)	(20.031)
House size	3.445	3.798	11.288**
	(4.258)	(4.308)	(4.876)
Farm equipment	-4.553	-5.472	-17.449
	(16.668)	(16.938)	(15.927)
Social capital	4.351	4.985	-15.134
	(7.368)	(7.469)	(9.267)
Financial difficulty	7.415	6.869	20.541
	(30.855)	(30.860)	(42.583)
Natural/biological disaster	16.514	17.768	21.743
	(20.339)	(20.427)	(27.108)
Constant	15.470	11.264	48.980
	(49.221)	(47.971)	(72.494)
Observations	235	235	153
R-squared	0.034	0.041	0.201

Notes: This table presents the results of a post-treatment regression specification to assess treatment effects, and the heterogenous treatments effects by gender (Female) and by total savings at baseline (Total savings (Bsl.)) on PPT pension savings (recorded in Ghana cedis). Controls consist of the variables that were unbalanced at baseline (Female; Age; Migration status; Farm equipment; Social capital; Financial difficulty; Natural/biological disaster), the attrition predictor (Salience of old age), and additional predictors from the Heckman selection model (Saving for old age; House size). See Table 3 for variable definitions. Huber-White robust standard errors are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

F Appendix: Robustness check psychological outcome variables; results for both short- and long-term

To check the robustness of our results for the psychological outcome variables, we use a LASSO double selection procedure (Belloni et al., 2014, 2016) as an alternative estimation approach. This data-driven procedure helps select covariates with non-negligible associations with the outcome variable and the treatment variables, and is designed to address a common challenge in empirical work, which is, selecting relevant covariates from a large set of potential controls.

We consider 40 variables (see Appendix Table F.1) covering personal characteristics, socio-economic characteristics, financial status, psychological features, and farming activities of respondents as potential controls. In addition, baseline values of the respective outcome variables are included in the LASSO procedure and are counted among the controls in Appendix Table F.2.

We aim to estimate a regression model of the following general form:

$$Y_{i,t=2} = \alpha X_{i,t=0} + \beta T_i + \epsilon_i. \tag{F1}$$

Here, $Y_{i,t=2}$ is the outcome variable during endline, $X_{i,t=0}$ represents a vector of independent baseline variables as controls (also presented in Table F.1) that can influence the outcome variable and T_i is the treatment variable and ϵ_i is an error term.

LASSO double-selection procedure is to select appropriate $X_{i,0}$s to use as control variables from the larger set of available control variables. The implementation follows three steps.

- First, run a LASSO regression of $Y_{i,2}$ on $X_{i,0}$ and collect the selected controls.
- Second, run LASSO regressions of each treatment variable in T_i on $X_{i,0}$ and collect the selected controls from these treatment equations.
- Last, estimate by ordinary least squares a regression of $Y_{i,2}$ on T_i and the union of the controls selected in the outcome and treatment steps above.

The intuition behind the procedure is as follows: Controls that are predictive of the outcome, selected in step 1, improve model fit and may help capture relevant factors affecting the outcome; while controls that are predictive of the treatment, selected in step 2, capture variables related to treatment assignment and thus potentially important sources of confounding. By including the union of controls selected in both steps 1 and 2, the double LASSO procedure reduces the risk that variables important for estimating the treatment effect are omitted from the final specification.

Our results remain robust to using this procedure. The estimated significant treatment effects are not substantially altered (see Table F.2). The results also remain robust when we repeat ANCOVA specification with the same (smaller) sample as in the Lasso specification (see Table F.3).

Table F.1: Descriptive statistics of all controls used in Lasso specification (except those reported in Table 3)

Variables	N	Mean	SD	Min	Max
Number of children	723	2.737	1.988	0	9
Educational level of children	723	1.573	1.61	0	7
Bank account	723	0.578	0.494	0	1
Non-bank savings account	723	0.657	0.475	0	1
Mobile money account	723	0.668	0.471	0	1
Use mobile money services	723	0.815	0.389	0	1
SUSU savings	723	0.444	0.497	0	1
Frequency of SUSU savings	723	0.385	0.487	0	1
Health insurance	723	0.935	0.247	0	1
Life insurance	723	0.015	0.122	0	1
Give money to social capital	723	1.124	1.033	0	5
Amount given to social capital	723	101.683	315.815	0	5150
Receive money from social capital	723	0.414	0.709	0	5
Amount received from social capital	723	35.061	143.587	0	2500
Money invested in cocoa farm	708	2360.484	2404.993	10	13980
Time invested in cocoa farm	711	156.127	136.729	2	1028
Plan for old age	723	0.835	0.371	0	1
High proportion of income on food expenditure	724	0.555	0.497	0	1
Financial satisfaction in old age (desired)	723	9.36	1.213	3	10
Financial satisfaction in old age (expected, no adjustment)	723	5.761	2.929	1	10
Financial satisfaction in old age (expected, with adjustment)	723	8.982	1.511	1	10

Notes: The table presents descriptive statistics for the potential control variables considered in the LASSO double-selection procedure. Number of children = number of sons, daughters, and children-in-law in the household; Educational level of children = average highest level of education attained by sons and daughters in the household; Bank account = 1 if respondent or household member has a bank savings account; Non-bank savings account = 1 if respondent or household member has a non-bank savings account; Mobile money account = 1 if respondent has a mobile money account; Use mobile money services = 1 if respondent uses mobile money services; SUSU savings = 1 if respondent or household member saves with a SUSU scheme; Frequency of SUSU savings = 1 if SUSU scheme requires periodic saving; Health insurance = 1 if household has health insurance; Life insurance = 1 if household has life insurance; Give money to social capital = number of categories of social capital to which respondent gave money, goods, or services in the past 3 months; Amount given to social capital = total value of money, goods, or services given to social capital in the past 3 months; Receive money from social capital = number of categories of social capital from which respondent received money, goods, or services in the past 3 months; Amount received from social capital = total value of money, goods, or services received from social capital in the past 3 months; Money invested in cocoa farm = total amount spent on agricultural activities in the past year, including input supplies and hired labor across all cocoa farm activities; Time invested in cocoa farm = total time spent on agricultural activities in the past year, including own labor and hired labor across all cocoa farm activities; Plan for old age = 1 if respondent reports thinking about the period when unable to work due to old age; High proportion of income on food expenditure = 1 if household's share of income spent on food in the past month is above the sample average; Financial satisfaction in old age (desired) = self-reported desired level of financial satisfaction in old age, measured on a 10-point scale; Financial satisfaction in old age (expected, no adjustment) = self-reported expected level of financial satisfaction in old age without changing current preparation, measured on a 10-point scale; Financial satisfaction in old age (expected, with adjustment) = self-reported expected level of financial satisfaction in old age with adjustments to financial preparation, measured on a 10-point scale.

Table F.2: Short- and long-run treatment effects on psychological outcome variables - LASSO

Variables	(1) Aspirations (Mdl.)	(2) Aspirations (Endl.)	(3) Hope (Endl.)	(4) Agency (Endl.)	(5) Pathways (Endl.)
Video	0.157**				
	(0.078)				
Vid. + Plac. Cal.		0.039	0.161**	0.166**	0.078
		(0.086)	(0.074)	(0.080)	(0.069)
Vid. + Sav. Cal.		0.103	0.117*	0.096	0.078
		(0.087)	(0.068)	(0.076)	(0.069)
Selected variables					
Aspirations (Bsl.)	0.475				
Hope (Bsl.)			0.095		
Agency (Bsl.)				0.138	
Financial satisfaction in old age (expected, with adjustment)			0.069		
Salience of old age			0.259		
Total farm size			0.005	0.004	
Elec. and electr. assets			0.062	0.075	
Number of baseline controls	41	41	41	41	41
Number of variables selected with non-negligible effect	1	0	5	3	0
Constant	0.145	-0.431	-1.167	-0.233	-0.273
Observations	634	548	548	548	548

Notes: The table presents the results of an OLS specification to assess treatment effects on short-term and long-term aspirations, as well as long-term hope, agency, and pathways, controlling for variables selected by the LASSO procedure (see Appendix F for details). For short term aspirations (Aspirations (Mdl.)) the procedure selects Aspirations (Bsl.). For long term aspirations (Aspirations (Endl.)) and pathways (Pathways (Endl.)) the procedure selects none of the controls. For long term hope (Hope (Endl.)) the procedure selects Hope (Bsl.), Financial satisfaction in old age (expected, with adjustment), Salience of old age, Total farm size, and Elec. and electr. assets. For long term agency (Agency (Endl.)) the procedure selects Agency (Bsl.), Total farm size, and Elec. and electr. assets. See Appendix Table F.1 for variable definitions. Huber-White robust standard errors are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table F.3: Short- and long-run treatment effects on psychological outcome variables - ANCOVA

Variables	Aspirations (Mdl.)		Aspirations (Endl.)		Hope (Endl.)		Agency (Endl.)		Pathways (Endl.)	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Video	0.157**	0.133*								
	(0.078)	(0.077)								
Video + Plac. Cal.			0.039	0.041	0.157**	0.147*	0.166**	0.152*	0.076	0.075
			(0.086)	(0.087)	(0.077)	(0.076)	(0.082)	(0.081)	(0.069)	(0.069)
Video + Sav. Cal.			0.112	0.109	0.092	0.083	0.088	0.069	0.070	0.075
			(0.087)	(0.087)	(0.071)	(0.070)	(0.078)	(0.077)	(0.069)	(0.069)
Female		0.053		-0.029	0.103		0.145*		-0.001	
		(0.095)		(0.096)	(0.079)		(0.076)		(0.082)	
Age		-0.003		0.004	-0.002		-0.002		-0.002	
		(0.003)		(0.003)	(0.002)		(0.003)		(0.002)	
Migration status		-0.098		0.023	0.018		-0.062		0.126**	
		(0.069)		(0.076)	(0.068)		(0.074)		(0.062)	
Salience of old age		0.102		0.233***	0.257***		0.244***		0.187***	
		(0.078)		(0.082)	(0.071)		(0.080)		(0.064)	
Farm equipment		-0.065		0.014	0.113**		0.121***		0.061	
		(0.063)		(0.055)	(0.045)		(0.046)		(0.046)	
Social capital		0.009		0.082***	0.017		0.013		0.028	
		(0.026)		(0.032)	(0.025)		(0.026)		(0.025)	
Financial difficulty		-0.036		-0.064	0.038		0.022		0.059	
		(0.084)		(0.091)	(0.080)		(0.080)		(0.083)	
Natural/biological disaster		-0.159*		-0.060	-0.037		-0.067		0.007	
		(0.084)		(0.075)	(0.062)		(0.067)		(0.061)	
Hope (Bsl.)					0.141***	0.128***				
					(0.035)	(0.034)				
Agency (Bsl.)							0.154***	0.146***		
							(0.032)	(0.032)		
Pathways (Bsl.)									0.036	0.024
									(0.027)	(0.026)
Constant	0.039	0.269	-0.483***	-1.040***	0.179***	-0.426**	-0.033	-0.238	-0.326***	-0.522***
	(0.068)	(0.261)	(0.061)	(0.210)	(0.053)	(0.177)	(0.061)	(0.194)	(0.046)	(0.154)
Observations	634	634	548	548	548	548	548	548	548	548
R-squared	0.202	0.217	0.003	0.033	0.041	0.078	0.055	0.092	0.006	0.034

Notes: : The table presents the results of an ANCOVA specification to assess treatment effects on aspirations, hope, agency, and pathways. For each outcome variable, the ANCOVA estimation is conducted on the same (restricted) sample used for the corresponding LASSO estimates in Appendix Table F.2. The treatment variables are video plus placebo calendar (Vid. + Plac. Cal.) and video plus savings calendar (Vid. + Sav. Cal.). Baseline controls consist of the variables that were unbalanced at baseline (Female; Age; Migration status; Farm equipment; Social capital; Financial difficulty; Natural/biological disaster) and the attrition predictor (Salience of old age). See Table 3 for variable definitions. Huber-White robust standard errors are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.